

Universidade do Minho
Escola de Engenharia

Diogo Poeta Miguens

Extraordinary transparency in continuous metallic thin films

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Extraordinary transparency in continuous metallic thin films

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STATEMENT OF INTEGRITY

I hereby declare having conducted this academic work with integrity. I confirm that I have not used plagiarism or any form of undue use of information or falsification of results along the process leading to its elaboration.

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ABSTRACT

Two dimensional materials have emerged as a toolbox for the fields of nanophotonics and nano-optoelectronics, providing new properties that are not possible to find in conventional materials. In particular, semiconductor transition metal dichalcogenides (TMDC) show unique optical properties.

This project proposes a nanostructure based on TMDC which achieves extraordinary transmission levels through a continuous thin metallic film.

We made a design of a metasurface of TMDC, placed both on top and under the continuous metal film, obtaining a final geometry TMDC-metal-TMDC. Due to the anisotropic high refractive index of the TMDC metasurface, photonic-plasmonic resonances are created, which makes it possible to obtain high transmission through the continuous metallic film. By means, of finite-difference time-domain (FDTD) simulations, we engender configurations showing up to 90% transmission for a 30 nm thick metal film of silver, which is an enhancement ten times higher than the transmission of the metal without the nanostructure.

This high transmission is tunable by means of the variation of the geometric parameter of the metasurface. This effect could be suitable for actual applications in optoelectronic devices, such as: transparent solar cell electrodes, plasmonic filters, polarizers, or to increase the efficiency of resonant cavities for photodetectors.

Keywords: Lithography, Transparent electrodes, Metal, Plasmon Resonances, TMDC.

SUMÁRIO

Materiais bidimensionais vêm a surgir como uma caixa de ferramentas para os ramos da nanofotónica e nano-optoelectrónica, fornecendo novas propriedades que não são possíveis de encontrar em materiais convencionais. Em particular, os "transition metal dichalcogenides" semicondutores (TMDC) apresentam propriedades óticas únicas.

Este projeto propõe uma nanoestrutura baseada em "TMDC" capaz de atingir elevados níveis de transmissão através de um filme fino de metal contínuo.

Fizemos o design de uma metassuperfície de "TMDC", colocada em cima e em baixo do filme metálico contínuo, obtendo-se uma geometria final "TMDC" -metal- "TMDC". Devido ao elevado índice de refração anisotrópico da metassuperfície de "TMDC", criam-se ressonâncias fotónicas de plasma o que permite obter uma elevada transmissão através do filme metálico contínuo. Através de simulações de "finite-difference time-domain" ("FDTD"), engendrámos configurações que mostram até 90% de transmissão para um filme fino metálico de prata de 30 nm de espessura, correspondendo a um aumento dez vezes maior do que a transmissão do metal sem a nanoestrutura.

Esta elevada transmissão pode ser adequadamente ajustada fazendo-se variar o parâmetro geométrico da metassuperfície. Este efeito pode ser útil em aplicações atuais para dispositivos optoelectrónicos, tais como: elétrodos de células solares transparentes, filtros de plasma, polarizadores, ou para aumentar a eficiência das cavidades de ressonância em fotodetetores.

Palavras-chave: Litografia, Elétrodos transparentes, Metal, Ressonâncias plasmónicas, "TMDC".

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List of Abbreviations

2D	Two dimensional
3D	Three dimensional
Ag	Silver
Au	Gold
conf	Configuration
EM	Electromagnetic
ff	Filling factor
IR	infrared
kV	Kilovolt
mat	Material
MoS_2	Molybdenum disulfide
$MoSe_2$	Molybdenum diselenide
mW	Milliwatt
Pa	Pascal
PC	Photonic Crystal
PDMS	Polydimethylsiloxane
pol	Polarization
SiO_2	Silica
SPP	Surface plasmon polaritons
SPR	Surface plasmon resonance
t	Thickness
T	Transmission
t_1	Thickness of the TMDC
t_2	Thickness of the Silica
t_metal	Thickness of the metal
TMDC	Transition metal dichalcogenide
t_TMDC	Thickness of the TMDC
UV	Ultraviolet
WS_2	Tungsten disulfide

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To my grandparents, who give me courage to fight no matter the circumstances in life. Through the doors along this path, I deeply know they are somewhere cheerfully happy for me.

Chapter 1: INTRODUCTION AND BACKGROUND

1.1 General Introduction to nanophotonics

One can't say the word "nano" without first having in mind how limited the possibilities are, even though there is a lot of research and knowledge on this area, we are still far from understanding and being able to manipulate this structures, maybe in some years this what we call "the future of technology" will no longer be the future but the past, there might be scientists working at even smaller structures after we comprehend and explore more of our molecular surroundings.

Think of two different walls that are in a same room, one wall looks flat and smooth, while the other is looking rough, you have this perception with your eyes that as you get closer to the walls you start to see how rough both walls actually are. By far, one is looking smooth while the other rough, which does not occur when you look closely, and this is exactly the same thing that happens when you study a material that looks smooth at the eye scale (macroscale) and how different it actually is when you observe it at smaller scales.

At macroscale, one has to think of such parameters like the contact angles on a 2D interface, at microscale the roughness details on both 2D and 3D surfaces where kinetic effects occur, but at nanoscale it's about the molecular heterogeneity and 3D interfaces while having in account thermodynamic and dynamic effects.



Figure 1.1: Photograph of Prof. Richard Feynman (1918-1988). (Image in the public domain).

In further advance, nanotechnology is the science, engineering and technology taken at the nanoscale, which is about 1 to 100 nanometers, but also, that exists at quotidian range of almost everybody. One nanometer is $1 \times e^{-9}$ (a billionth) of a meter, impossible to see with the naked eye and

only in the early 80's (preciously in 1981 with the invention of the scanning tunneling microscope and in 1985 with the atomic force microscope) became possible to study and develop nanotechnology, coming along the possibility to measure intermolecular forces and to see atoms.

Nanophotonics or nano-optics is part of the nanotechnology that studies the behavior of light on nanometer scales, as well as interactions of nanometer-sized objects with light [1].

There was this lecture "There's Plenty of Room at the Bottom", that happened once in 1959, dictated by Richard Feynman (**Figure 1.1**) that made the following questions: "What could we do with layered structures with just the right layers? What would the properties of materials be if we could really arrange the atoms the way we want them?" [2]. This physicist, father of nanotechnology, was a visionary as he predicted the advances of nanotechnology 50 years ahead of its time, since we are now able to manipulate the atomic structures and this made it possible to create new materials, to replicate advanced bio-inspired technology but also to understand how we can improve the already existent ones in our daily. For example: nanostructures forming structural colors that can be seen in nature on scales of butterflies or some feathers of birds, this happens because those structures scatter the light which makes the waves of certain frequencies constructively interfere [3], and the nanostructure create the color against the formation of colors due to absorption and reflection of different wavelengths (**Figure 1.2)**, this phenomena can be used to avoid the loss of colors in cars [4].

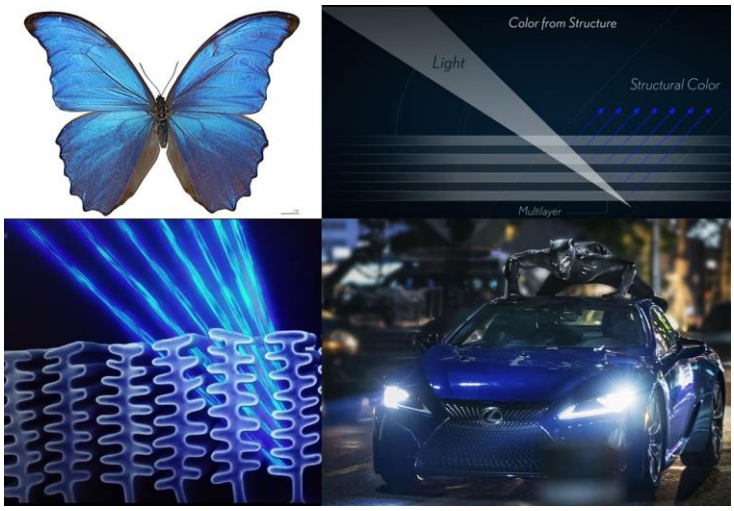


Figure 1.2: Structural color: From nature structural color to industrial applications (composition).

In photonics there are different regimes in which the light interacts with the material due to an effective medium. These interactions light-matter depend on the ratio of the wavelength (λ) with the geometrical feature. For example, in the case of optical gratings (**Figure 1.3**) this interaction depends on the grating period (Λ) and the wavelength (λ). When the $\Lambda > \lambda$ it operates in the diffraction regime, in

the middle it lies an unexplored regime: the near-wavelength regime, where the Λ is between the λ of the grating material. Finally, the deep-subwavelength regime when the $\Lambda < \lambda$.

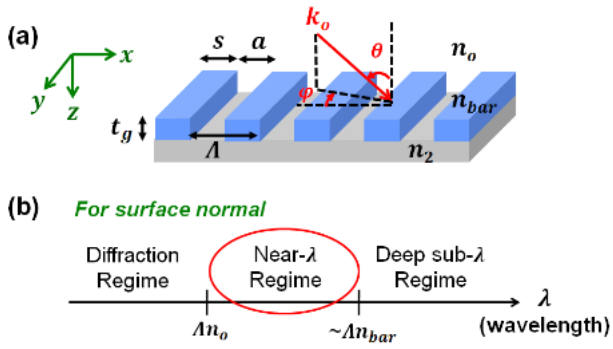


Figure 1.3: Regimes in optical gratings.

1.2 Motivation: The urge to have ultrathin high conductive transparent electrodes

Indium tin oxide (ITO) is one of the most widely used transparent conducting oxides, because of its electrical conductivity and optical transparency, however as indium is scarce and ITO is limited in flexibility due to its ceramic structure, many studies have been dedicated to new transparent conductive electrodes (**Figure 1.4**).

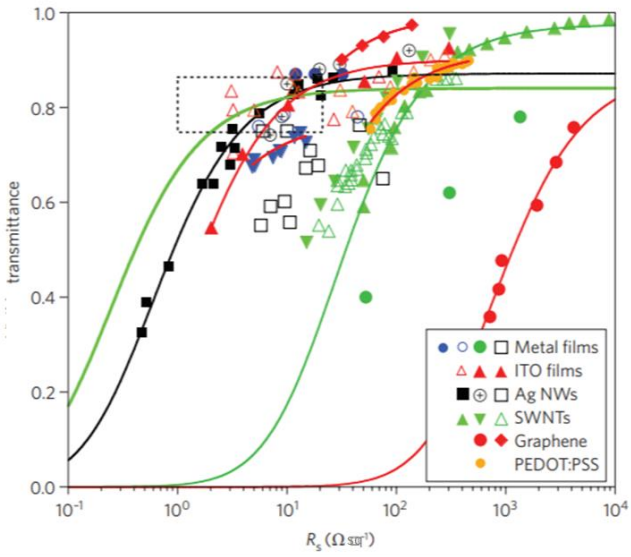


Figure 1.4: Transmittance and sheet resistance for different transparent electrodes [38].

A flexible transparent electrode (FTE) is one of the most essential parts for the next generation of flexible optoelectronic devices, including solar cells, displays, and solid-state lighting devices. Although a lot of candidate materials for the FTE, such as metallic nanowires, carbon nanotube, and graphene have

been investigated, each material has its fundamental limits in FTE applications, such as low transmittance, high sheet resistance and rough surface morphology.

In order to be considered for industrial application, a transparent electrode material should show a sheet resistance less than 100 ohm⁄sq at more than 80% transmittance [5]. However, the exact requirements on the electrode performance depend strongly on each specific application.

For the fabrication of OLEDs, for instance, the decisive criterion is a high conductivity,[6] whereas the integration in organic photovoltaic (OPV) devices requires very high transparency.[7] Or, as the main motivation of this work, the design and fabrication of an ultrathin transparent electrode which can be very important and useful in tandem solar cells (**Figure 1.5**), due to the fact that it reflects for the visible and transmits for wavelengths beyond 1000 nm, which makes it useful, for example in a tandem solar cell comprised of a perovskite front cell and a colloidal quantum dot back cell.

Also, the device is tunable, making it possible to change the wavelength depending on what the user requires, allowing the system to be used as a polarizer in the infrared (IR) or telecommunications range.

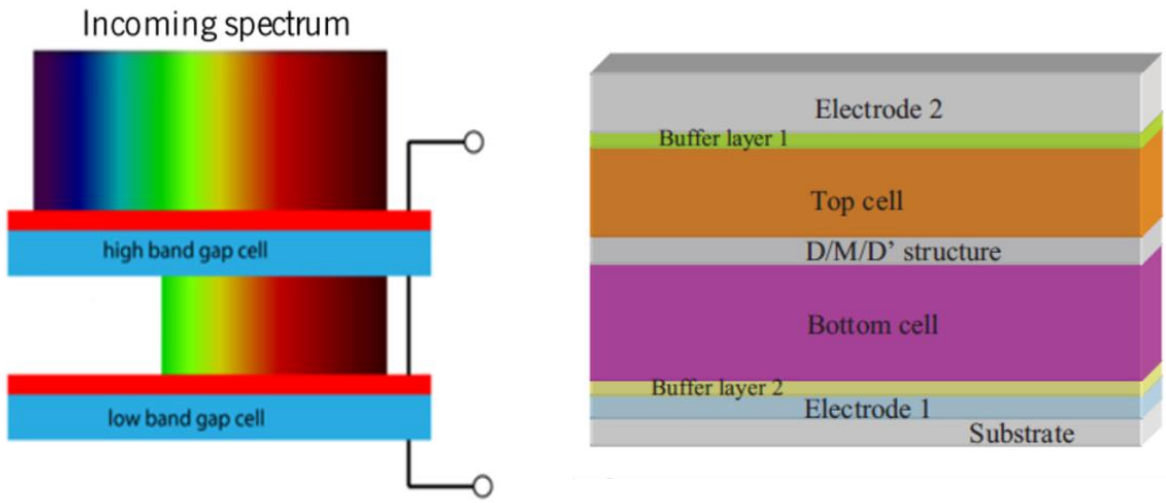


Figure 1.5: Scheme of tandem solar cell. Dielectric/metal/dielectric medium electrode in a tandem solar cell.

Dielectric/metal/dielectric (DMD) electrode structure is a promising candidate for next-generation flexible transparent electrodes. When a metal mirror layer is embedded between two dielectric layers, the dielectric/metal/dielectric tri-layer film could suppress the reflection from the metal in the visible range and achieve a selective transparent effect.

With this, we propose the use of a transition metal dichalcogenide (TMDC) as a dielectric, they belong to the family of 2D materials and are ideal to use in the fabrication of transparent electrode systems, since they have a very high refractive index.

1.3 2D materials

Properties of the materials are largely dependent on the size and morphology, as a consequence the control over the synthetic methodologies became an issue. This is because the growth of the materials in nanoscale is largely dependent on the thermodynamic and kinetic barriers in the reaction as defined by the reaction trajectory and influenced by vacancies, defects, and surface reconstructions. The simultaneous control of particle size and shape together with their uniformity is one of the key objectives in many synthetic procedures.

Atomically thin 2D materials interface, thickness and defects are determined by their performance. Recent advances in the development of atomically thin layers of van der Walls bonded solids have opened up new possibilities for the exploration of 2D physics as wells as for materials for applications. Like it's discussed in the following section, those materials are tantalizing when it comes to its applications on the optoelectronics field, for photodetectors, electronics, but also for valleytronics (control over the valley degree freedom, a local maximum/minimum on the valence band) [8].

About 2D materials, when we change from macro to nano scale, things as we know, might work differently, so one has to be ready to whatever changes he/she may encounter when studying the molecular structure of a material.

It is important to notice that in this work the relevant ones are TMDCs. There is plenty of work and research on this area, maybe because graphene is included in here and over the last years it was a top tier research due to its new stupendous discoveries, but there are plenty of other 2D materials, like for example: phosphorene, germanene, or even also the TMTC's (transition-metal trichalcogenide), such as: $\mathrm{TiS_3}$(titanium trisulfide), or $\mathrm{Zr}S_3$(zirconium trisulfide) [9].

1.4 Transition metal dichalcogenides

Transition metal dichalcogenide (TMD or TMDC) are inside the family of two-dimensional (2D) materials with a lot of potential applications, due to their magnetism, optical, electronic and electric properties.

The structure of the TMDC's is very similar to graphite, which is based on layers held by weak interactions of Van der Waals forces Their formula is: MX_2, where "M" stands for transition metals, "X" for chalcogenides and "2" for "di" (from Greek di-, shortened form of dis "twice," which is related to duo "two" [10]). They can be called of monolayers, triple layers or sandwiches depending on the context. In the structure shown in the **Figure 1.6**, each layer consists of three atomic planes with a usual thickness

of six to seven angstroms with a hexagonal plane of metal atoms sandwiched between two planes of chalcogen [11].

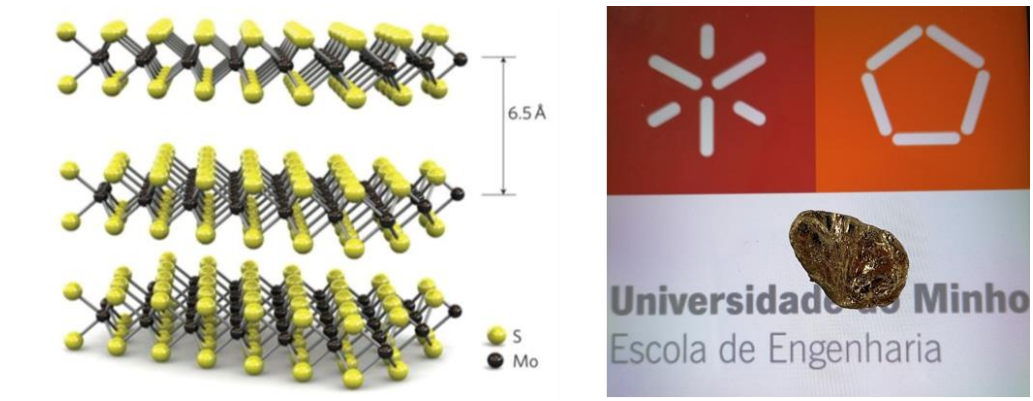


Figure 1.6: Typical atomic structure of a transition metal dichalcogenide and mineral of MoS2 used in this work.

Some of the semiconducting TMDCs are composed for the transition metals: Mo (molybdenum) or W (tungsten) and coupled to two chalcogen atoms such as S (sulfur), Se (Selenium) or Te (tellurium). These materials are known for having large exciton binding energies, leading to strong light-mater interactions and short radiative lifetimes.

They can be modified at nanoscale by plasma processing because of its effective processes, it is easy to be controlled and also the energy efficiency, it has effect over: phase transition, etching, surface passivation, doping, defects engineering and hetero structuring. Some of its applications are: hydrogen generation, lithium-ion batteries, and gas sensors [12].

Different from other bulk TMDC crystals, the following materials are direct band semiconductors and according to their properties and applications, we opted to work with: MoS_2 (molybdenum disulfide), $MoSe_2$ (molybdenum diselenide) and WS_2 (tungsten disulfide). Our focus on the TMDC's centers in their optical properties, we want to create a structure that makes the metal become transparent and for that, it's required the analyses of the optical parameters.

1.4.1 Optical properties of TMDC

It is believed that besides this Lorentz model, there are also proof of other resonances as double resonance and multi resonance. The Lorentz model helps to understand the optical response of dispersive materials. Charges in materials are treated as harmonic oscillators.

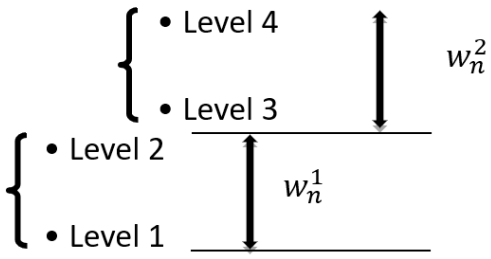


Figure 1.7: Illustration of Lorentz-Model with 4 transition levels.

For each transmission between levels one and two we introduce a harmonic oscillator, which contributes for the response of the optical field.

If the frequency of light is closer to the transitions level 1 and level 2, then only this transition would matter, this happens because we are in the presence of a resonance effect, which makes this transition the strongest one. The atomic polarizability is given by the equation:

$$\alpha(\omega) = \frac{e^2}{m}\sum \frac{f_{n0}}{\omega_{n0}^2 - \omega^2 - i\omega\delta_{n0}} \tag{1.1}$$

It is known that the closer the frequency to the resonance the higher the amplitude of the response, also the sharper the resonance the higher the quality of the image obtained will be. So, when we are interested in optical properties of a material in different range of wavelengths we have to consider that this material is dispersive that means that the refractive index depends on the wavelength of work.

The refractive index is described by many as how slow the light travels in one medium compared to another (reference medium), and it's calculated by the following formula:

$$\mathrm{n} = \frac{c}{v} \tag{1.2}$$

where n = refractive index;
c = velocity of light of a given wavelength in empty space;
v = velocity in the medium.

It can be named of absolute refractive index or relative refractive index depending on if we are comparing it to vacuum or to another medium, correspondingly. For absorbing media, the refractive index could be represented as a complex number $\hat{n} = n + i\kappa$ where κ is the extinction coefficient. This last coefficient is related to the absorption of the material.

1.4.2 Refractive index

Refractive indexes of TMDCs were determined in literature by reflectance measurements of the samples at room temperature (**Figure 1.8**). For the TMDC, the two highest wavelength peaks in the reflectance spectra correspond to the excitonic features associated with interband transitions [13].

These semiconducting TMDCs present a high real part of the refractive index which is interesting compare with other materials (**Table 1.1**) in order to show that it is one of the highest in nature. This phenomenon makes that we have a high optical contrast with respect other dielectrics as air making strong resonance in nanostructures as we see in next sections

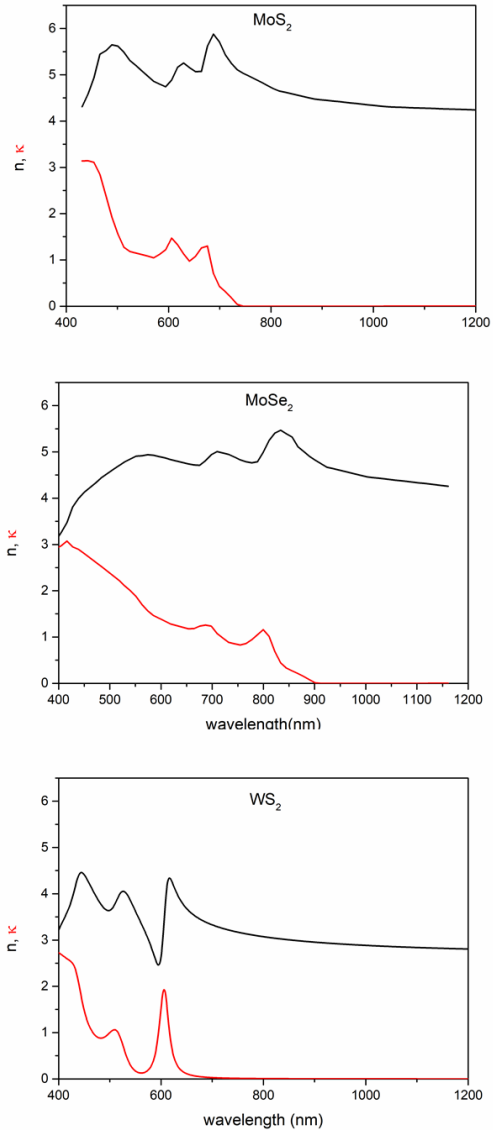


Figure 1.8: Refractive index and absorption coefficient for MoS2, MoSe2 and WS2.

As we can see in the **Figure 1.8**, these TMDCs show a very high real part of the refractive index, being lower than 4 for WS_2. If we focus in the coefficient of absorption, then the MoS_2 and WS_2 have very low absorption for wavelengths higher than 700 nm and $MoSe_2$ has no absorption for wavelengths higher

than 900 nm. These two criteria: high real parts and low absorption, make MoS_2 a good candidate for the design of a dielectric metal dielectric transparent electrode.

Table 1.1: Refractive index and absorption coefficient for MoS2, MoSe2 and WS2.

Material	Refractive index (real part)
Wavelength 500nm	
Air	1.00
Glass	1.52
Silver (Ag)	0.05
Gold (Au)	0.97
SiO_2	1.46
MoS_2	5.62
$MoSe_2$	4.57
WS_2	4.56

Finally, although it is not represented here, we had it in account for the simulations, which is the anisotropic refractive index component.

When we have, for example, a 30nm thin metal film we have no transmission and all the wave is reflecting, but when we place gratings of TMDC top and down the continuous layer of metal we create high resonance, which creates high transmission.

1.4.3 Photoluminescence

When light is emitted from a material in its relaxation process after being absorbed from it, we name this process of photoluminescence (PL) and it depends on the energy and momentum of the incoming photons, the size of the energy gap and the shape of the energy gaps to be effective. The result of incoherent microscopic polarization and coherent polarization induced by the pump pulse, is the cause of the PL source. Due to the presence of excitons, there is an enhancement on the absorbance of single layers, that in turn also improves the absorbance at the visible part in the electromagnetic spectrum.

In solids or molecules, incoming photons can be absorbed and make an electron rise from the valence band to the conduction band. The semiconductor TMDC's energy gaps ranges from one to two

electron volts. When the band gap transitions from and indirect to a direct gap, there is a decrease on the number of layers and the photoluminescence yield increases [14].

TMDCs exhibit unique optical properties that evolve from the surface effects that arise during the transition of an indirect bandgap to a direct bandgap when bulk materials are scaled down to monolayers (**Figure 1.9**).

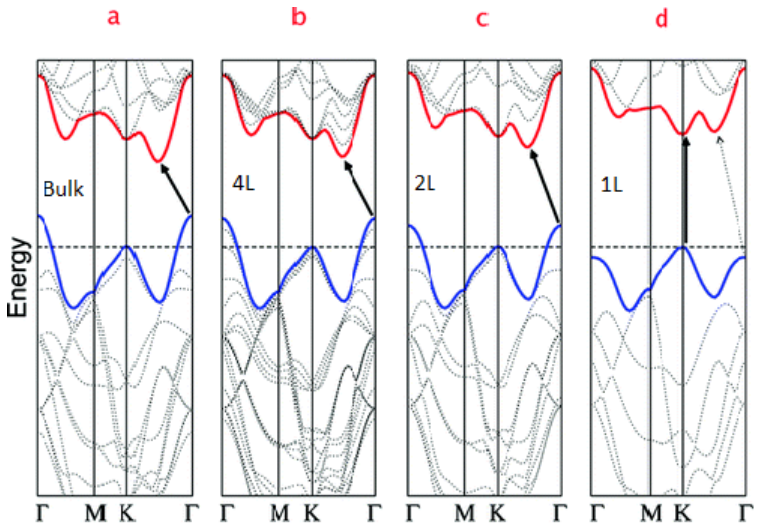


Figure 1.9: Example of calculated MoS2 bands for (a) Bulk, (b) 4 layers, (c) 2 layers and (d) monolayer [15].

The tunable bandgap in TMDCs is also accompanied by strong PL and large exciton binding energy, making them a potential candidate for a variety of optoelectronic devices, including solar cells, photodetectors, light-emitting diodes, and phototransistors.

While still in the bulk phase and even though the gap changes with the number of layers and becomes direct in the monolayer limit, this gap is initially indirect, being the gap indirect not all the phonons will be able to provide the necessary wavevector to enable the absorption of the photon and creation of an electron-hole pair The direct gap produces a higher PL(**Figure 1.10**), this means that it is not necessary for phonons to be created or absorbed in the process of light absorption, which makes the process way more effective than in its counterparts with more layers [16].

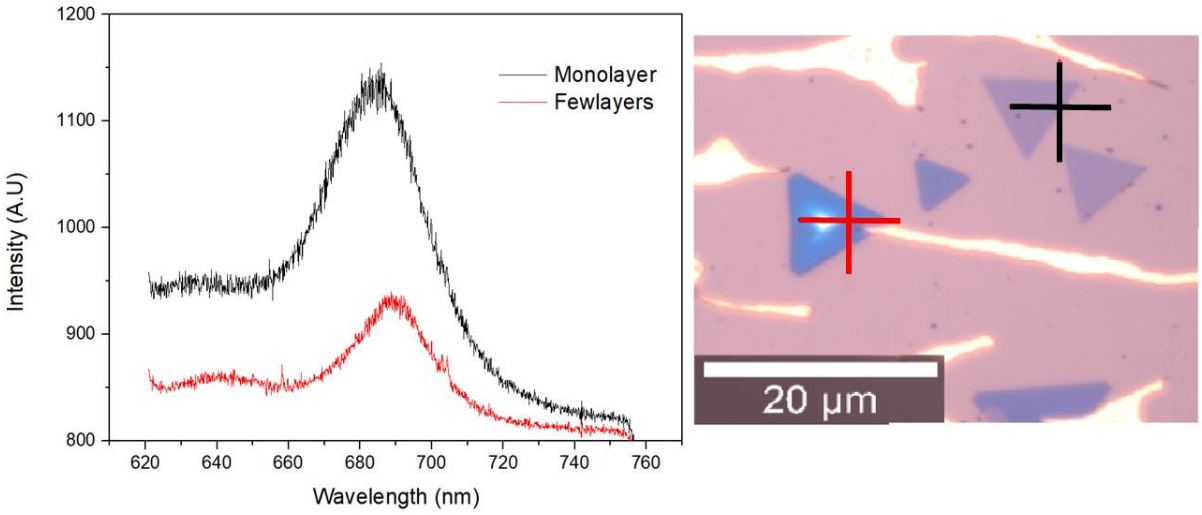


Figure 1.10: (a) Measurements of MoS2 photoluminescence. (b) Optical image showing the regions where we measured PL.

1.4.4 Raman Spectrum

This technique was used to characterize the chemical structure of the materials. In TMDCs their Raman spectra can be used to check the quality of the materials (**Figure 1.11**) and to estimate the thickness of the thin films looking the position of the vibration peaks and the intensity of those (**Figure 1.12**).

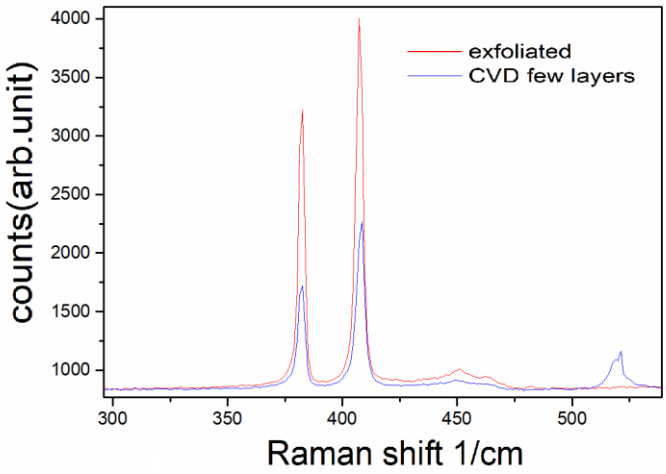


Figure 1.11: MoS2 Raman signature comparison between an exfoliated bulk and a few layers CVD showing the good quality of the samples.

In the following chapter 2, we describe in more detail how Raman spectroscopy works.

1.5 Optical properties of metals

Having the perfect conditions, high temperature and really low particle density, which can be individual conditions or also happen simultaneously, is what makes it possible to have what we call: plasma, which is nothing more than an ionized electrical conducting gas of charged particles. This effect can be seen in northern lights (also called "auroras"), where in the last year, 2020, scientists from the National Institute of Polar Research, in Japan, published an article based on their work of optical calibrations to ensure the measurement of the emission intensity caused by auroras, which is usually weak and for that it is necessary to have a high sensitivity calibrated instrument to measure the radiation caused by this this phenomenon [17].

Plasma exists in the ionosphere (80 to 640 kilometers above Earth's surface) due to the ultraviolet (UV) lights from the Sun that ionize air molecules, I'd suggest the reader to check reference [18], if it's of one's curiosity to see how spectacular this phenomenon is, visually seen from space on what scientists call "airglow", it is an article of Nasa about ionosphere discoveries, last updated on December 11, 2019, written by Lina Tran, which I found their results to be really interesting.

One main effect of a plasma is the plasma oscillation, caused by the motion of electrons and ionized nuclei, where there is a Coulomb's force that tries to restore them to their original positions, which is the reason of this oscillations that resonate at a certain frequency, known as the plasma frequency, given by the following equation:

$$\omega_P^2 = \frac{N_e^2}{\varepsilon_o . m} \tag{1.2}$$

where ω_p = is the plasma frequency of the material;
ε_0 = constant of electric permittivity of vacuum.

In 1956, Pines studied energy losses experienced by fast electrons travelling through metals, which he attributed to collective oscillations (which can be controlled by adjusting the spatial region filled by electron plasmon), calling them of 'plasmons'. When those are confined to thin metal films, couple with photons, resulting in the formation of surface plasmon polariton (SPP) waves, corresponding to electromagnetic excitations. It was also in 1956, that the word 'polariton' was introduced by Fano, to describe the coupled oscillation of bound electrons and light inside transparent media.

I also find relevant to make, about the surface plasmon effects and its advantages for renewable energy applications on metal surfaces and nanoparticles, a comparison between a statement we found on the results obtained on chapter 3, that we obtained better transmission for smaller thickness of the materials used in the simulations, and according to a paper about improved photovoltaic devices, written

by Atwater Harry and Polman Albert, with a very small reduction of the thickness layer of silicon there is a reduce in the material cost per output power [19]. This is important again, on how smaller scales can make such a difference, most of its times being positive and worthy of being studied.

We intended to create plasma, has was already stated, so that our structure could have resonance effects in order to increase the transmission, it is known that plasmas and metals to a certain extent are wholly reflective for electromagnetic waves of frequencies below the medium's plasma frequency, and totally transmissive for frequencies above it.

1.5.1 DOS

The free-electron theory of metals describes that electrons in any solid are able to move freely in that entirely solid, because the nuclei of individual metal atoms hold the outer electrons by very weak electron forces. Although, the electrons can freely roam in the metal, they can't escape the surface of the metal, in fact there is a low probably that the electron will gain enough energy to escape to the surrounding atmosphere. Since electrons are trapped within the metal, we can think of the metal as a finite well, in this case electrons have a potential energy of zero, but we know there is a really low probability that the electrons will jump to higher values at the edge of the metal.

Quantum mechanics explain how electrons have both wave and particle-like properties, while tunneling is a wavelike nature effect. Called of quantum tunneling, to refer to when electrons move through a barrier they shouldn't be able to, now if we integrate quantum physics in this subject, we know that an electron trapped in a potential well (metal) has quanta (plural of 'quantum') sized energy states. Thus, by the Pauli exclusion principle, all electrons must be given by a certain unique set of quantum numbers and only two electrons can be found in any given electron orbital. In solid metals there is a large number of electrons, therefore there is also a large number of possible energy states.

In order to describe all the different quantum states that the electron can take inside our metal, we define a mathematical function known as the density of states (DOS), which gives the distribution of all the possible quantum states per unit volume per unit energy that the electrons in the metal can take.

$$\rho(E) = E^{1/2} \times \frac{8\sqrt{2}\pi m^{3/2}}{h^3} \tag{1.3}$$

where
- ρ = density of states
- m = mass of electron
- E = energy of the electron
- h = Planck's constant

It is also possible to calculate the number of quantum states (n) per unit volume (V), by integrating the function of Eq. (1.3) and multiplying it by the volume of the metal solid. Doing this, we obtain this integral function for two energy values,

$$n = \int_{E_1}^{E_2} \rho(E) dE \times V \tag{1.4}$$

where n = number of density states
V = volume of the solid

SPPs feature flat dispersion characteristics with wavelength much shorter than the free space propagating waves, resulting in large density of states at select frequencies corresponding to the excitation of SPP modes.

1.5.2 Metal thin films

The metal layer for flexible transparent electrodes should be thin, for high transmittance and continuous for low resistivity. Silver, copper and gold are the main candidates in the thin metal film sandwiched between two TMDCs nanostructures.

In this work, we choose in the design of the transparent nanostructure silver because show the less transmission and high conductivity. Copper whose electrical and optical properties are very close to those of silver, and whose price is largely lower could be used in these systems. For a chemical stability, due to silver oxidizes easily; gold is used in the experimental proof of concept. So, although gold is "semitransparent" in the visible range in the near infrared show low transmission and it will be useful to show the concept of extraordinary transparency. These phenomena are shown in **Figure 1.12** which shows the transmission of a 30 nm thick and thin film for different metals.

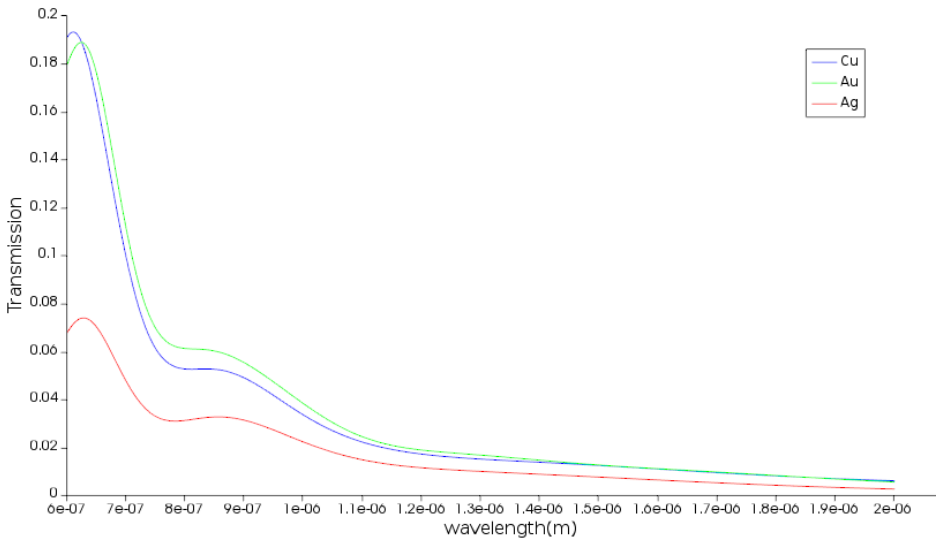


Figure 1.12: Transmission of a thin film 30 nm thick of copper (blue line), gold (green line) and silver (red line).

1.6 Photonic and plasmonic resonances

In order to overcome the low interaction of light with these ultrathin films, where a small thickness is necessary for having a flexible device, nanostructures creating photonic and plasmonic resonances will be necessary to increase the interaction and the transmission in the transparent electrode. So, we will design systems as gratings or photonic crystals (PCs).

Photonic crystals (PCs) are artificial media with a periodic index contrast. An electromagnetic wave in a photonic crystal is described by a band structure, which relates the frequency of nodes to the wavevectors, the fundamental properties of nodes are: scale invariance and orthogonality.

The low band is the dielectric band, and the high band corresponds to the air band, for high dielectric structure we will use transition metal dichalcogenides. So, PCs are composed of periodic dielectric nanostructures, which are design with the intent to propagate electromagnetic waves (EM), they are used to modify or control the flow of light [20].

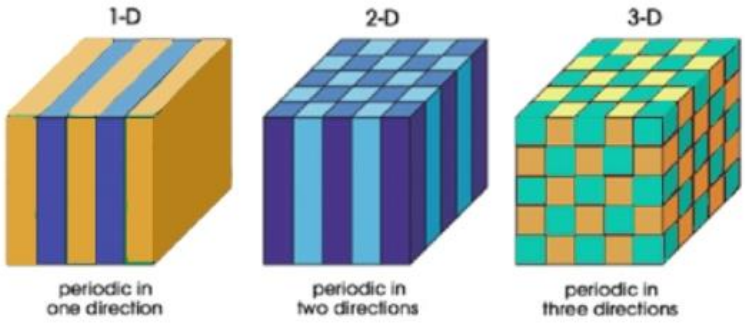


Figure 1.13: Photonic crystals in 1D, 2D and 3D [21].

PCs may have a novel use in optical data transmission but are not extremely prominent. They may be used to filter for interference in a fiber-optic cable or increase the quality of the transmission. In addition, they can be used to divide different wavelengths of light. PCs can already be manufactured at close to the nanoscale.

The light trapping effect is taken in consideration for the applications of our structure, as it can be useful to have an enhancement over the transmission for solar cell devices, this effect is necessary to couple photons with wavelengths higher than 600 nanometers [22].

Scattering is the general name for light "bouncing" off of some object in its path but to describe its phenomenon its preferable [23,24]. Some examples of scattering are seen in **Figure 1.14**. The actual scattering event can consist of absorption, emission, and propagation away from the object. In general,

the process may be elastic or inelastic, although, except for plasmonics, we will generally limit our discussion to elastic processes.

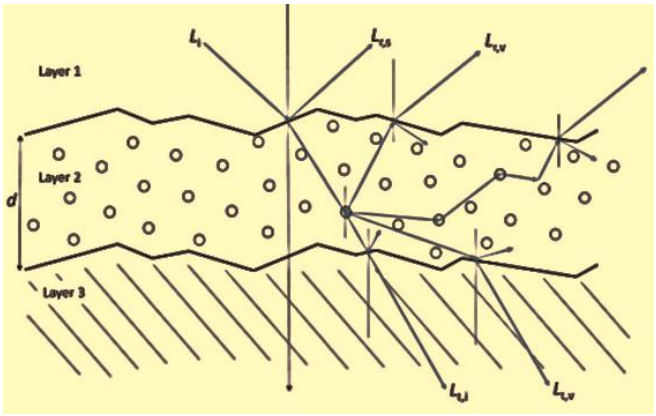


Figure 1.14: Scheme of scattering and optical paths.

For most materials, the emission following absorption involves an electron relaxation process in the scatterer, which takes place in less than 10−8 s. Usually, this emission time is so short that its granularity is not considered in light trapping.

1.7 Photonic resonances

A resonance occurs due to the motion of the low frequency of the atoms and the high frequency of the electrons.

It was predicted that the system would create resonance, in the simulations we verified to have the following resonances which are explained a little in detail in the next paragraphs.

1.7.1 Thin film Fabry-Pérot interference

In a thin film Fabry-Pérot interference, due to the presence of parallel reflecting surfaces and this being the reason why these interferences occur, light can either be transmitted or reflected. Reflection can occur in upper and lower boundaries between different materials, and it can also happen interference between reflected waves because of the difference in the optical path (Δx). For a certain wavelength (λ) the conditions are:

$$\Delta x_{const} = m\lambda \tag{1.5}$$

$$\Delta x_{dest} = (2m + 1)\frac{\lambda}{2} \tag{1.6}$$

where (1.5) stands for total constructive and (1.6) for total destructive interference conditions.

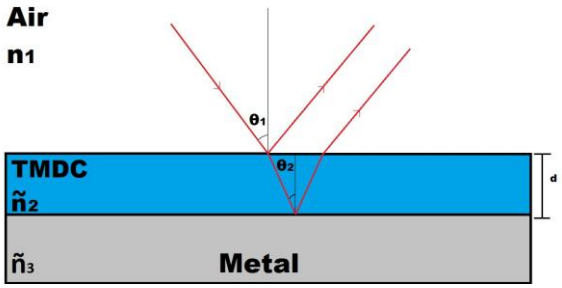


Figure 1.15: Scheme of an interference occurring in a structure with three mediums: air, metal and TMDC.

Δx is dependent on the refractive indexes of both media, on the angles of incidences for both surfaces $\theta 1$ and $\theta 2$ and on the thickness of the TMDC film. One very common application of Fabry-Pérot interference is the quarter-wave coating, where there is a dielectric of thickness, d, positioned between a medium with a low refractive index, n1, (such as air) and a medium with higher refractive index ,n3, in a way which the light reflected on the interface air-dielectric will obtain a phase of $\Delta\delta 1 = \pi$ and light reflected on the second interface will increase its phase by $\Delta\delta 2 = 2kn3d+\pi$, corresponding to the phase gained by traveling a distance 2d inside the film with refractive index n3. The trick is to make the thickness d be equal to $\frac{\lambda}{4}$, so the phase difference between the two waves as they interfere in air will be as given in the equation 1.7.

$$\Delta\delta_{total} = \Delta\delta_2 - \Delta\delta_1 = (2kn_3\frac{\lambda}{4} + \pi) - (\pi) = kn_3\frac{\lambda}{2} = \frac{2\pi n_3}{\lambda_0}\frac{\lambda_0}{2n_3} = \pi \quad (1.7)$$

where $\lambda 0$ is the wavelength of the incident wave in air. So, light will interfere destructively in air, meaning that there will be no reflection.

In the **Figure 1.15**, we see that the medium 3 corresponds to metals, here we can select the metal for the bottom material, having in consideration the complex refractive indexes and complex reflection coefficients, although at the end, the result is the same: quarter-wave dielectric leads total destructive interference outside (in air). In order to have total absorption in thinner films, which is possible to achieve due to the destructive interference, a metal with higher penetration depth can be used.

1.7.2 Mie Resonance

The Mie resonances of dielectric inclusions provide a novel mechanism for the creation of magnetic or electric resonance based on displacement currents and offer a simpler and more versatile route for the fabrication of isotropic metamaterials operating at higher frequencies. One of the important concepts of dielectric Mie-resonant nanophotonics is associated with the Kerker effect that drives the

unidirectional scattering of light from nanoparticles and metasurfaces. This effect will be very important to obtain high transmission in the nanostructures based on dielectric/metal dielectric.

1.7.3 Diffracted modes

Diffraction occurs when light faces obstacles, such as defects or apertures, but for this to happen, the following relation has to be obeyed (for diffraction in the same medium):

$$\mathrm{m}\lambda_0 = \mathrm{nL}(\sin\theta_m - \sin\varphi) \tag{1.8}$$

m	=	order of diffraction;
λ_0	=	wavelength of the incident electromagnetic wave
n	=	refraction index of the medium
φ	=	incidence angle relative to the normal
L	=	periodicity of the obstacles/defects
θ_m	=	diffraction angle relative to each order m

It can produce interference patterns due to optical path differences between wave fronts (and thus a phase difference), such as what happens in Bragg diffraction.

Equation (1.8) can also be rewritten as:

$$\mathrm{n}K_0\sin\theta_m = \mathrm{n}K_0 sin\emptyset + \frac{2m\pi}{L} \tag{1.9}$$

If the wave is impinging normally, we have:

$$\mathrm{n}\mathrm{K}_0\sin\theta_\mathrm{m} = \mathrm{K}_0\frac{2\pi\mathrm{m}}{\mathrm{L}} \tag{1.10}$$

Besides telling us the angles for which diffraction occurs, the equations above also give insights about momentum: K0 is the free space wavevector $\mathrm{K}_0 = \frac{2\pi}{\lambda_0}$ associated with the incoming wave. Before scattering, the parallel wavevector is:

$$\mathrm{K}\|= \mathrm{nK}_0\sin\emptyset \tag{1.11}$$

After scattering, according to (1.10), the new parallel component of the wavevector can be written as:

$$\mathrm{K}\|= \frac{2\pi\mathrm{m}}{\mathrm{L}} + \mathrm{nK}_0\sin\emptyset \tag{1.12}$$

Therefore, when interacting with the periodic structure of the medium, the parallel momentum of the wave is enhanced by: $\pm\frac{2\pi\mathrm{m}}{\mathrm{L}}$, which implies a change in momentum, even if its original parallel momentum is zero. The periodic structure only exchanges reciprocal lattice vectors $\vec{\mathrm{G}} = \frac{2\pi}{\mathrm{L}}$, but instead

of L being the crystalline lattice parameter, it can be the distance between slits or holes of the gratings, meaning that an increase in momentum can only be determined by the array periodicity.

It's of particular interest to use this possibility of giving parallel momentum to light in light trapping [25], since there is a chance to obtain guided modes, plasmonic modes and Bloch modes.

1.7.4 Guided mode resonance

The resonance created by propagating waves that are strictly connected inside a certain material (waveguide), which are also surrounded by another material with lower refractive index, is known as guided mode resonance. This is possible because according to Snell's law, total internal reflection can occur if the angle of incidence θi has a value greater than the critical angle $\theta_c = \arctan(\frac{n_{external}}{n_{internal}})$.

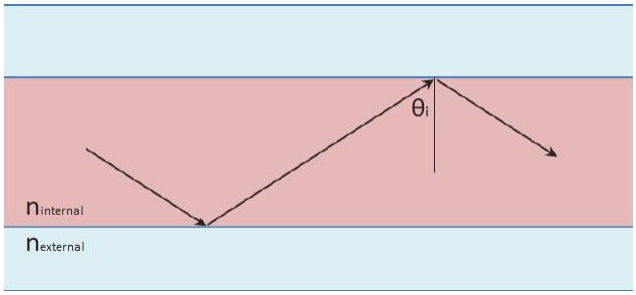


Figure 1.16: Above the critical angle, total internal reflection is possible.

Inside a material, guided modes can provide a more effective absorption, since light stays longer inside the medium, this creates more electron-hole pairs and therefore it generates more photocurrent.

1.7.5 Surface plasmon resonances

The incoming electromagnetic radiation can possibly cause plasma oscillations of the free electrons in any metal, such oscillations can reradiate for certain incoming frequencies.

Their use advantageously changes the scattering at the front and reflection at the back of a structure [25,26]. The plasma oscillations occurring in the metal cause energy dissipation, because of the collision of electrons within the plasma, due to this, the use of plasmonic behavior at front meshes and at back metal interfaces has its relevant importance [22], however it requires a precise control over the collisional losses in front and back structures.

From a periodic array of nanostructures, which can be nanoparticles or gratings, at a front and back structure of a metal, there are also two other effects happening due to the periodic arrays that collect resonances, which are a result of the cooperative and radiating coupling of the localized surface plasmon

resonances, one based in the hybridization (referred to as surface plasmon polaritons) of the trapped traveling modes in thin dielectric nanostructured periodic metal arrays [27] and the second effect, which happens because of the periodic structure in the plane of the array causing diffracted orders of the reradiated energy [26].

In light trapping and absorption, this cooperative and radiation coupled modes can exist in the absorber and be advantageous for this approaches, as they can be used in solar cells [22].

The huge number of free electrons in a metal are induced to oscillate around the positively charged cores of metal atoms due to the incoming electromagnetic waves. Those oscillations are quantized, and the quanta are called plasmons. It is possible to induce such plasmons in dielectric-metal interfaces who are called SPPs. To have an example on how the oscillations, in this case the oscillation involving both charge motion around nuclei in the metal and electromagnetic waves in the dielectric, are guided along the interface, we can compare it to how light can be guided by an optical fiber, since the basis of the system is similar.

A metal real and imaginary parts of the permittivity vary with the frequency of the impinging electromagnetic wave, and this directly affects the behavior of the surface wave.

Chapter 2: EXPERIMENTAL AND THEORETICAL METHODS

2.1 Introduction

Before we head onto the design based on simulation procedures and the fabrication of the structure, this chapter contains the most relevant techniques and methods used in this thesis.

There might be some attempts by other scientists on trying to fabricate a structure similar to ours, but as far as we searched we might be the first ones trying to create a metasurface of TMDC top and down a continuous metallic thin film, so no doubt we need many attempts and ideas on how to fabricate such a device, where the high transmission will occur depending on the positions where the flakes are localized, in other words, there must be a flake of TMDC on top and another flake of TMDC under the thin metal layer exactly on the same position (which is a very difficult condition to achieve since we are working in very small scales).

The synthesis of 2D TMDCs can be broadly classified as top-down and bottom-up.

2.2 Deposition methods

2.2.1 Exfoliation:

The exfoliation method is by far the most practical and simplest method on the whole list of this chapter when compared with the other methods, we press very lightly the blue tape on the MoS_2 flake (**Figure 2.1(a)**) in order to take a very small flake of the sample. After that, since we want a monolayer, we make the process that can be seen in the **Figure 2.1(b)**, which is called the scotch tape technique and basically, one needs to press the flake removed from the sample and exfoliate along the tape, until the tape is filled with thinner and smaller size flakes.

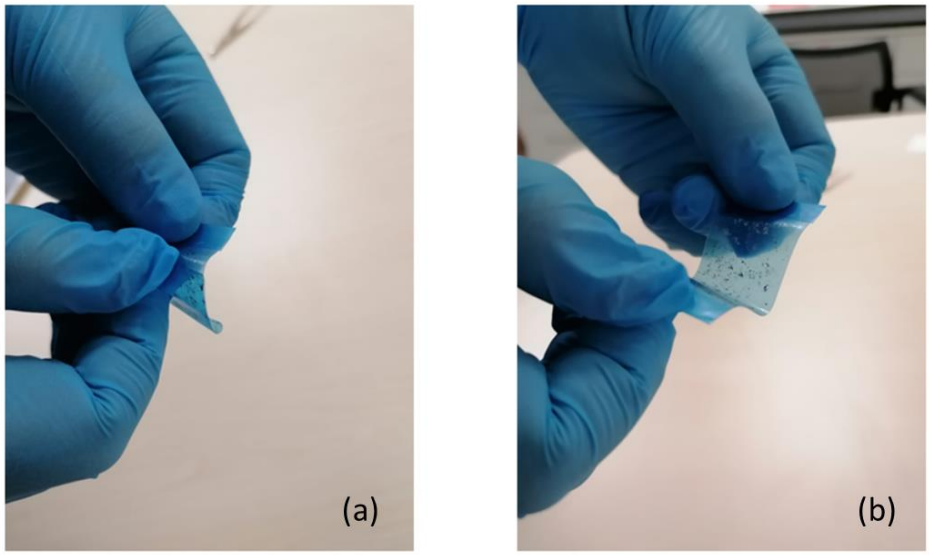


Figure 2.1: Scotch tape technique.

By doing the mechanical exfoliation, one can produce 2D materials of high quality. With the scotch tape technique, the lateral size of 2D materials is limited as they tend to break during exfoliation. The scotch tape technique has a low yield and is limited to laboratory usage. In the top-down approach, van der Waals forces between individual layers are broken, referred to as exfoliation.

We also need to have in attention the use of gloves and be as careful as possible in order to not contaminate the flakes.

2.2.2 Chemical vapor deposition:

Chemical vapor deposition (CVD) is a bottom-up deposition technique used to synthesize high-quality 2D material on a large scale. CVD is a vacuum deposition technique in which a solid material is deposited from a vapor phase precursor by a chemical reaction [28]. With this technique it is possible to do a growth of thin films, coatings, two-dimensional materials and powders. The process of CVD stars with a flow of the reaction gas mixture that has to pass through the coating surface, after it goes through the deposition process, the result of the fluid dynamics is kind of a motionless boundary layer which occurs in the vapor close to the substrate.

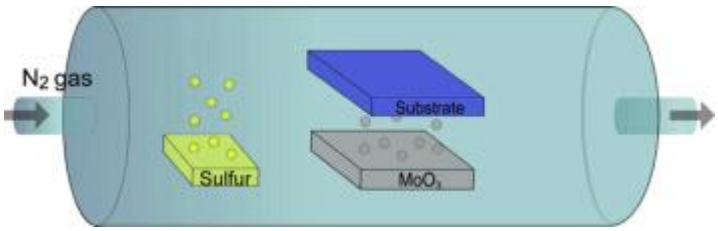


Figure 2.2: Principle of chemical vapor deposition [29].

As a family of deposition techniques, like the thermal CVD (which has its negative sides like the high-temperature deposition and limitation in the usage of several materials as substrates), plasma-enhanced CVD, photo-CVD, etc., those differ depending in how the energy is supplied to the system in order to activate the chemical reaction's precursor molecules. The byproducts required for the CVD process need to be neutralized and are also quite expensive.

2.2.3 Lithography:

Lithography (in Greek "Líthos"—stone; "graphein"—to write [30]) is a planographic printing technique using a plate or stone with a smooth surface, invented by Alois Senefelder in 1796 [31]. Nowadays, it's referred to the fabrication of one- and two- dimensional structures in which at least one of the lateral dimensions is in the nanometer range. Lithography replicates patterns, which can be positive and/or negative masks, into underlying substrates.

It is subdivided into photolithography, electron beam lithography, X-ray and extreme UV lithography, focused ion beam and neutral atomic beam lithography, soft lithography, colloidal lithography, nanoimprint lithography, scanning probe lithography, atomic force microscope nanolithography, and others. The etching process is top-down and the growth of nanolayers is bottom-up.

Bottom-up syntheses involves the assembly of small (atomic or molecular) units into the desired structure. The method to achieve this is to assemble the nano building blocks (NBBs) into a nanostructure with well-defined shape and structural design.

2.2.4 Laser ablation:

In this thesis we use laser ablation technique as a lithographic tool. We use a laser coupled to a Raman Witec setup mentioned in next section. We can move the scanning stage of the microscope sample holder by means of software codes provided by Witec. A series of handmade programs in Matlab were developed to translate to the Witec programmable commands and make all kind of patterns of lines. It is worth noting that by using this technique, the size of our structures is limited by the Rayleigh criterion:

$$\Delta w = \frac{0.61\lambda}{NA} \tag{2.1}$$

where Δw is the width of the line, λ is the wavelength of the laser, and NA is the numerical aperture of the objective. In our case λ is 532nm corresponding to a green laser, NA is 0.9, resulting in a width Δw ≈360nm.

Although electron beam and photolithography have been used extensively to pattern 2D materials, they suffer from high costs, complexity, vacuum operation requirements, and more importantly are prone to leave behind contaminates or polymer residues, causing damage or unwanted doping which can inadvertently degrade their electrical properties. In this regard, laser ablation is a promising technique to pattern 2D materials that is in situ, resist-free, and maskless. Laser ablation has been used in 2D materials, for example Park et al. demonstrated a procedure that involved laser ablation-based graphene patterning and subsequent transfer of the patterned graphene to an arbitrary substrate [32].

We made some lithography in the flakes of MoS_2, managing to do straight lines that were very thin in width but only possible on flakes with a very low number of layers, as can be seen in **Figure 2.3**.

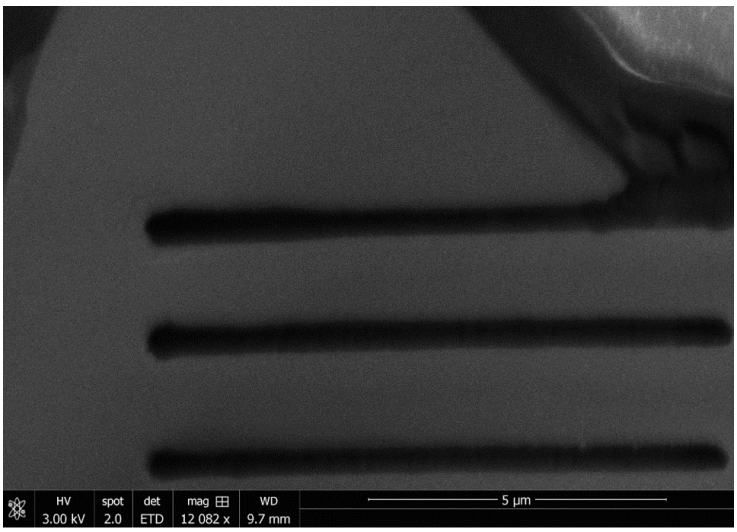


Figure 2.3: Lines made with Raman in a MoS2 flake.

The laser was not working properly but with some attempts it became useful in discovering which flakes had more layers and which did not, since their color and contrast change with the number of layers (or thickness) it has, and it was only possible to do lines in some flakes (pink and blue flakes) which we considered to be the ones with fewer layers.

2.3 Optical characterization

2.3.1 Raman:

The Raman is a type of absorption spectroscopy, which is non-destructive, with no sample preparation, and its acquisition is done within very few minutes.

With the help of Raman, it is possible to verify if the exfoliated samples prepared with flakes of MoS_2 contain gold and if it the sample is clean and good to use, the characterization gives immediate results on chemical analysis, physical structure, and characteristics spectral patterns of materials ("fingerprinting") to determine quantitatively and semi-qualitatively the sample [33].

The principle of Raman effect is based a sample exposed to monochromatic light in visible region, the sample absorbs light which in turn is transmitted to the sample. However, a small part of the light is scattered by the sample in all directions

Raman scattering is a two photon process, the electrons have different vibrational levels as can be seen in **Figure 2.4**, which is a diagram of the different levels of electrons that helps to understand this effect. When an incident monochromatic light interacts with an electron in the sample, the electron absorbs energy from the incident photon, rising to a virtual state of energy. The energy transferred is given by the formula: $E = hv_i$. The electron then falls back to an energy level by losing energy. If the energy lost is equal to the energy of incident photon ($hv_s = hv_i$), the electron falls back to its initial level emitting another photon. Since those energies are equal the least photon has the same frequency of the incident photon, occurring the Rayleigh scattering.

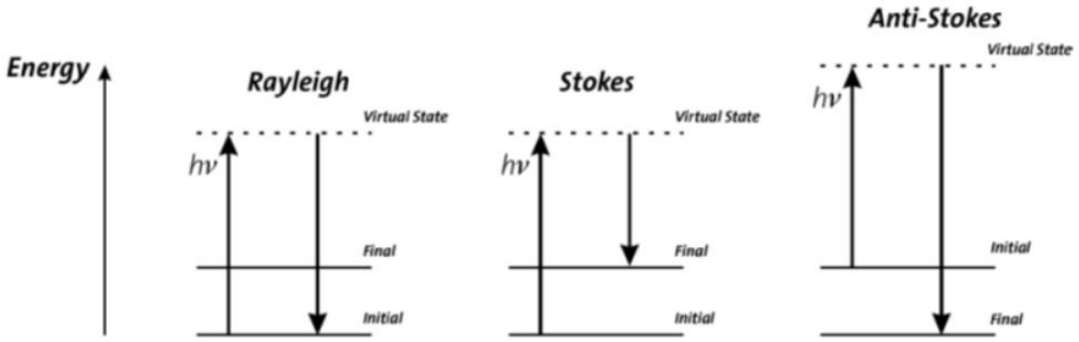


Figure 2.4: Scheme of Raman scattering.

However, sometimes electrons when losing energy from the virtual state can fall back to a different vibrational level, in this case the energy lost by the electron is different than the energy absorbed from the incident photon. As a result, the photon emitted by the electron has energy different from the incident photon, this is possible when the frequency of the emitted photon is different than the incident photon, this gives rise to Raman scattering. Depending upon the final energy of the electron or the final vibration level of electron, Raman scattering can be separated in two: stokes lines and anti-stokes lines. If the frequency of the scattered photon is less than the frequency of the incident photon, stokes lines are absorbed in Raman spectrum when the electron absorbs energy. Similarly, when frequency of emitted

photon is greater than the incident photon, anti-stokes lines are absorbed, in other words, the energy is released by the electron

In the laboratory at INL (International Iberian Nanotechnology Laboratory), we have the model Witec Alpha300R Confocal Raman (**Figure. 2.5**), which has three possible laser excitation wavelengths at 532, 633, and 785 nm. The Raman spectrometer is coupled to an optical microscope allowing the confocal setup to reach higher magnification of samples with spatial resolution in orders of sub-micron, but also with the confocal optics of this system is possible to analyze volume within the sample to be spatially filtered in both lateral (XY) and (Z) axes. An objective 100x with NA aperture 0.9 is and a green laser (532 nm) with power of 1mW and 20 mW is used for Raman and lithographic techniques, respectively.

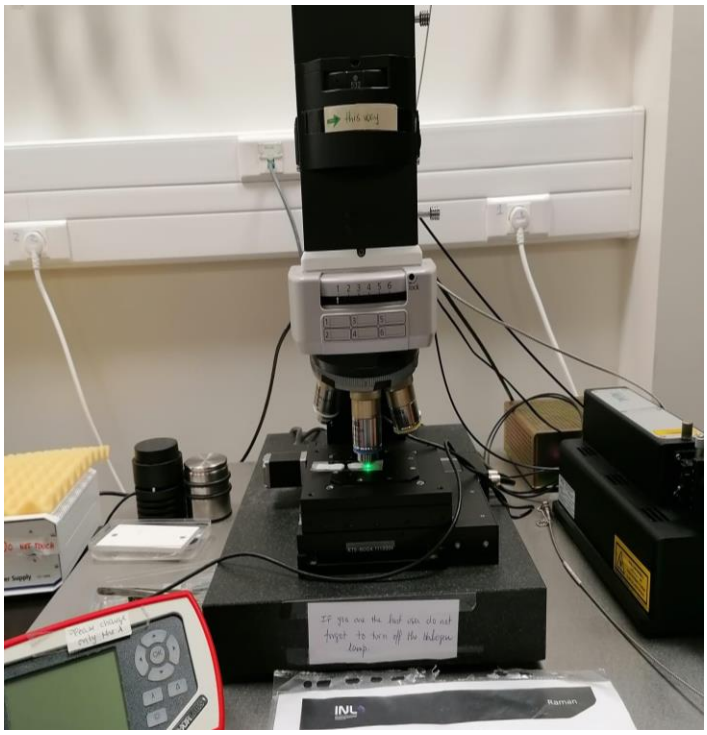


Figure 2.5: Witec Alpha300R Confocal Raman.

2.3.2 Scanning electron microscope (SEM):

A scanning electron microscope (SEM) is used to analyze and determine the samples for their microstructure, chemical composition, morphology, and surface topography.

This device produces the image by raster scanning the specimen with a focused beam of electrons. The electrons interact with the specimen generating various signals, including secondary electrons (SE), back-scattered electrons (BSE), characteristics x-rays, transmitted electrons, etc., as shown in **Figure 2.6**.

Each signal contains different information. The electron beam either passes through the specimen unaffected or experiences three different interactions with atoms in the specimen.

SEM used in this work is shown in **Figure 2.6** and the model is the FEI Quanta 650 FEG (corresponding to the device we used in the lab at INL), which is a very high versatile SEM possible to be used under high, low and extended vacuum conditions. This model operates with a Schottky field emission gun that provides high beam intensity and stability and allows for high resolutions.

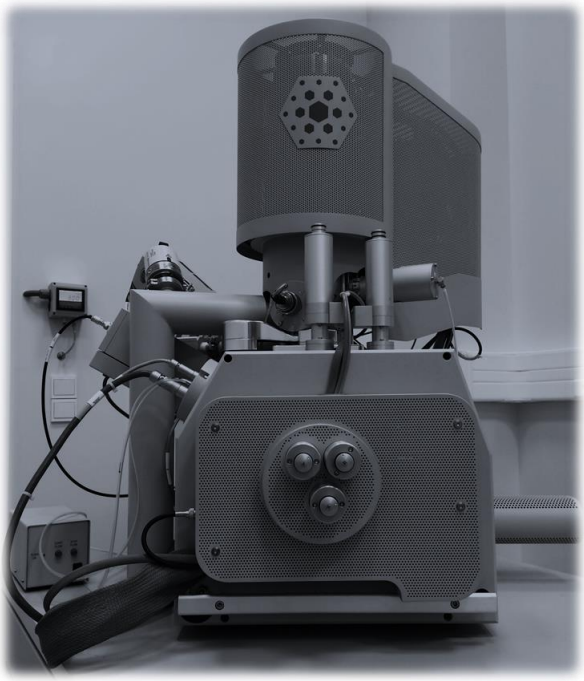


Figure 2.6: Scanning electronic microscope FEI Quanta 650 FEG.

The electron acceleration voltage ranges from 1 to 30 kV. Besides the large range of vacuum conditions (6e-4 – 4000 Pa) and acceleration voltages, the versatility of the instrument is further enhanced by a Peltier cooling stage and an abundance of detectors. Secondary electrons can be detected by an Everhardt Thornley SED (secondary electron detector) at high vacuum. Multiple samples can be loaded at the same time and the stage allows for x-y-z, rotational and tilting movements.
With the above listed equipment, the machine is routinely used to characterize conductive and non-conductive samples with secondary electron, back scattered electrons and EDS.

2.3.3 Ellipsometer:

The Accurion nanofilm_ep4 imaging ellipsometer (**Figure 2.7**) is a new generation of microscopic thin film, surface and materials metrology tool that combines auto nulling ellipsometry and

microscopy to enable surface characterization with lateral resolution as small as 1 micron. This enables resolving sample areas 1000 times smaller than most non-imaging ellipsometers, even if they use micro spot spectroscopic option.



Figure 2.7: Accurion nanofilm_ep4 imaging ellipsometer.

The Accurion nanofilm_ep4 spectral imaging ellipsometer is very useful for localized measurements of the optical properties of thin films and devices.

In this work we do not use the system as an ellipsometry device, but we use the device to measure the transmission with direct sample visualization by contrast image with a lateral resolution as small as 1 micron. For this purpose, we established the emission and collection arms in horizontal position and measured the transmission in the different areas of the sample for a wavelength range from 700 nm to 1000 nm. Unpolarized light is used with a 50x objective in the collection arm.

2.3.4 Fourier Transform Infrared spectroscopy:

We use Fourier Transform Infrared spectroscopy (FTIR) to measure the transmittance in the visible and near infrared with the FTIR from 600 to 2000nm and for this purpose, unpolarized light was used. The equipment used was a Bruker Vertex 70 FTIR spectrophotometer attached to an optical microscope (Bruker Hyperion) as shown in **Figure 2.8**. This allowed us to control visually the position in which we were taking the measurements. The spectra were recorded using a 50x objective with

numerical aperture (NA) of 0.55 for emission, and a 100x with NA 0.80 for collection. The transmission measurements were normalized against cleaned glass and PDMS transparent materials.



Figure 2.8: FTIR system coupled to a microscope.

2.4 Electrical characterization

The most common technique used for measuring sheet resistance is the four-probe method. This technique involves using four equally-spaced, co-linear probes (known as a four-point probe) to make electrical contact with the material. In this work we use a homemade four-point probe which uses sharp needles equally-spaced as probes (**Figure 2.9**).

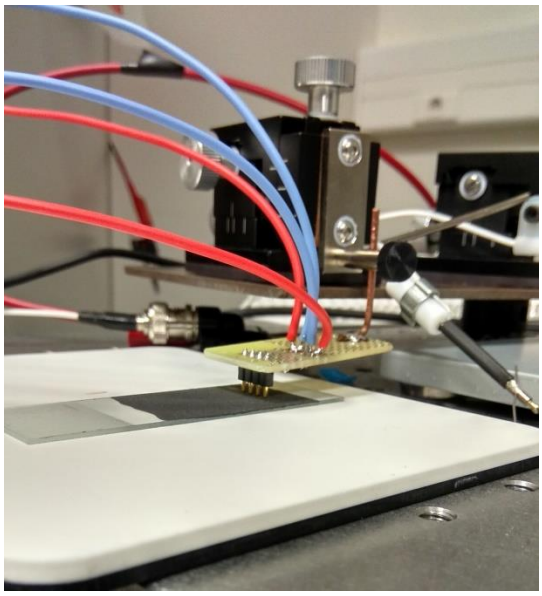


Figure 2.9: Homemade four probe point system.

This technique is also known as the Kelvin technique, a method of eliminating wire and contact resistances from a resistance measurement. **Figure 2.10 (a)** shows the circuit resistances of a four-point probe measurement.

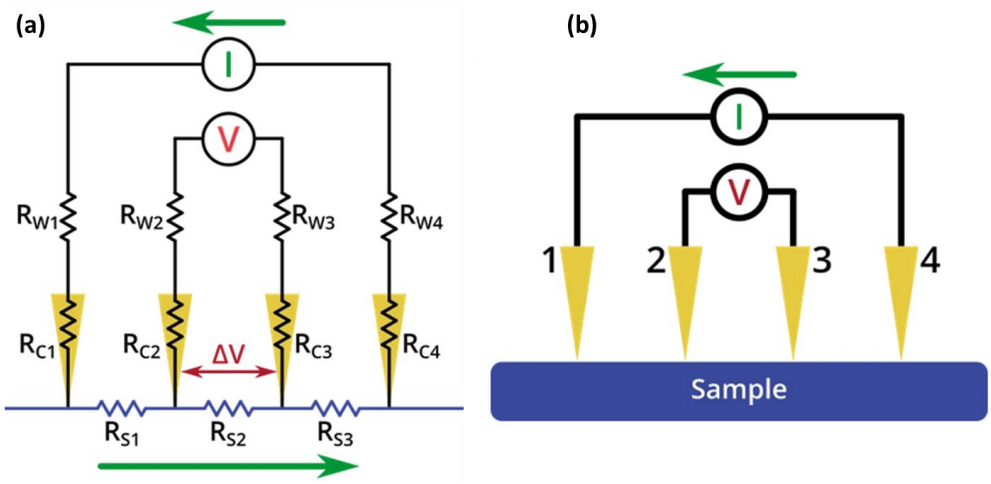


Figure 2.10 : Equivalent circuit diagram of a four-point probe, showing the wire resistances (RW), contact resistances (RC), and sample resistances (RS). The green arrows represent current flow [34].

As no current flows through the inner two probes, the wire resistances of R_{W2} and R_{W3} and the contact resistances of R_{C2} and R_{C3} do not affect the voltage measured at the voltmeter. This means that the measured decrease in voltage (ΔV) between the inner two probes arises entirely from R_{S2}. Therefore, ΔV can be used along with the applied current in the sheet resistance equation to calculate the value of R_{S2} (i.e., the sheet resistance).

So, finally, a DC current is applied between the outer two probes (1 and 4) and a voltage drop is measured between the inner two probes (2 and 3) (**Figure 2.10 (b)**). The sheet resistance can then be calculated using the following equation:

$$R_s = \frac{\pi}{\ln(2)} \frac{\Delta V}{I}$$

where, R_s is the sheet resistance, ΔV is the change in voltage measured between the inner probes, and I is the current applied between the outer probes. The sheet resistance is expressed with the units Ω/sq, or "ohms per square", to differentiate it from bulk resistance.

In addition to the factor of π/ln (2), a geometric correction factor is often required. The correction factor is based upon the size and shape of the sample, as well as the positioning of the probes, and accounts for the limitation of current pathways through it.

2.5 Numerical methods

2.5.1 Finite difference time domain method (FDTD):

To study the behavior of the transmission, we made several simulations using a commercial-grade by a computational program Lumerical Solutions, based on a finite-difference time-domain (FDTD) method, where FDTD is software that analyzes the interactions of UV, visible light and IR with complex structures.

This software uses a rectangular, Cartesian style mesh, which is already programmed to create an automatic smaller mesh in both higher index materials, also by doing this, a constant number of mesh points per wavelength is generated. Even though there is this default mesh, it is sometimes required to add another mesh, because the original default one isn't enough for complex structures where variations on a large field occurs.

But actually, FDTD (when not related with Lumerical) is a method that consists in explaining the effect of the air-filling factor on the optical properties of defect-free PCs [35], it is useful to calculate cross-sections at the nanometer scale wavelength, but there are also other advantages for this method:

i. Its numerical simplicity and straight forward physical basis since it is a numerical solution of Maxwell's equations.
ii. Its ability to make predictions that are accurate enough, so that the results can be used for real-world photonic designs.

After the simulation tests, we give use to the Broadband Fixed Angle Source Technique (BFAST) for anti-reflection coating, as basically one thing we want to do after we create a structure with high transmission, it's to study the depths of what has been simulated, to know where there are losses of reflection since we know we will create a 3D device and the behavior from 2D to 3D changes.

BFAST is a planewave type, very useful for simulating periodic structures, which have a broadband plane wave type source illuminated at an angle. There are other two types of planewave, which can be: Bloch/periodic or diffracting, although for our case the planewave type we want to use is the BFAST, since it's suitable for materials like metals that have such high losses.

Chapter 3: DESIGN CONFIGURATION AND SIMULATIONS

3.1 Introduction

It is very difficult to make a thin film of metal transparent due to the low penetration of the electric field inside it and the consequent high reflection.

In our case, we have a thin metal layer (30 nm thick) which has almost no transmission (T), and its absorption is almost zero and the metal is only reflecting due to its natural characteristics and physics of the material previously explained. When we add a high index (TMDC) and low index (air) material making a metasurface on top and bellow to the metal, we can obtain transmission for a determined wavelength (**Figure 3.1**), the layout of this structure can be seen in **Figure 3.1(a)**, where the structure is made of: TMDC-metal-TMDC.

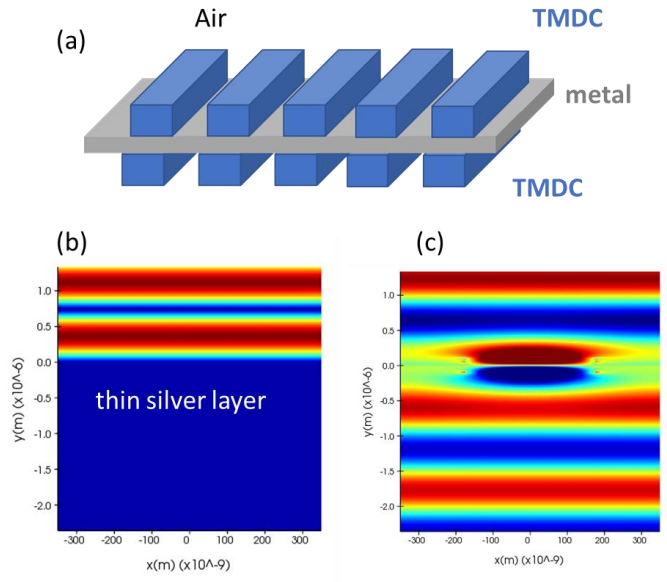


Figure 3.1: Scheme of the metasurface of our study. (b) High reflection and no transmission of an impinged plane wave from the top on a thin silver film. (c) Unit cell of the nanostructure TMDC/metal/TMDC showing high transmission.

Along this chapter we try to design a metasurface having the highest transmission possible, but also to have this high T propagating further and constant in the wavelength, as what is called of broadband T. Our first attempt was based on the paper of Wakabayashi [36] where they have a grating composed of an alternating constant high dielectric and lower proven that the SPR depends on the periodicity and the FPR on the thickness of the gratings.

So, we modify different geometrical parameters in the nanostructure as the thickness of the TMDC(t_TMDC) the period or the filling factor (ff). The ff is a dimensionless quantity and an adjustable parameter, that goes from 0 to 1, 0 meaning that there is only metal on the layer and 1 when we have the whole metal cover for a flat thin film of TMDC.

It happens that, by increasing the ff, the TMDCs nanostructures get closer to each other. Another way to understand this is by the following formula:

$$\mathrm{ff} = \frac{\text{width of the TMDC}}{\text{period}} \tag{3.1}$$

where the period is measured from half the TMDC to another half of the closest TMDC and when we decrease the period, the ff increases.

We made several simulations with different materials TMDCs, three different configurations, and different geometrical parameters, and many trials to check if the T is robust with the variation of the angle of incidence (in which we use BFAST technique), all in order to achieve high and broadband transmission. These attempts had the proposal of making transparent electrodes for solar tandem cell and devices suitable to be used as a polarizer in the infrared range and in telecommunications.

3.2 Materials and methods

For the materials we use silver (Ag) as the metal due to its high conductivity which is better for an electrode and the low transmission for the bare thin metal film and three TMDCs: WS_2, MoS_2 and $MoSe_2$.

We made two different configurations, so we could have more parameters to study and analyze, but also to discuss with the experts on the fabrication methods, which were possible to create in the laboratory. The first configuration has air involving the structure, a thin layer of metal and a grating of lines of TMDC deposited over and down the metal (**Figure 3.1 (a)**).

The importance of working with air as a component is the easiness to create holes in the structure (etching process).

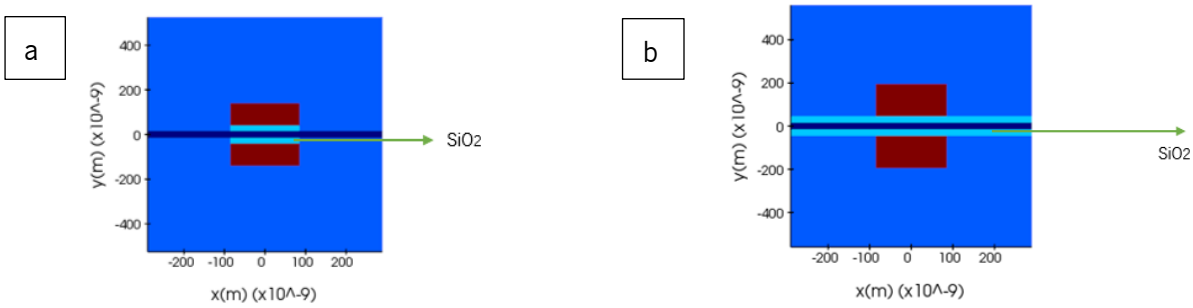


Figure 3.2: Index preview of the update made to the first configuration where we added silica. A - second configuration on 2D, where in blue is the air, in dark blue the metal, in red the TMDC and now in light blue the SiO2. B - Index preview of the 2nd configuration, the colors correspond to the same materials of A, where the difference is seen in the SiO2 that here is deposited all over the metal.

The second configuration has TMDC, air, silver and silica (SiO_2), the idea of adding SiO_2 is based on a paper in which the authors use silica as a low index dielectric with small thickness [37]. The purpose of adding SiO_2 to the structure is to create broadband transmission (to have T in more wavelength values).

In order to see what would happen in our case, the SiO_2 is deposited on the metal in two similar ways, as shown in **Figure 3.2 (a)** where the silica is deposited only below the TMDC without covering the rest of the metal that has no TMDC and **(b)** where the SiO_2 is placed above the thin metal layer.

Our next studies, which have focus on the second configuration of MoS_2 and the first configuration of $MoSe_2$, but in this case the lines will be changed the grating of lines by a squared array, and we study if it is better to have circular or hexagonal holes in the TMDC: for periods of 500 and 600 nm. Also, the manipulation of the system in order to achieve T for the telecommunications range, with a period of 700nm and a wavelength range from 1250 to 1650nm. In the results, it is better explained why this preference for the second configuration of MoS_2 with SiO_2, and the first configuration of $MoSe_2$ which has no silica. So, every time we speak of the second configuration, it refers to the following structure order: MoS_2-SiO_2-silver-SiO_2-MoS_2, and the first configuration is: $MoSe_2$-silver-$MoSe_2$.

Finally, a study of the electric fields, in lines, for the first configuration of $MoSe_2$ with 500nm period will be shown in order to understand the physics behind these systems.

The whole design process of these metasurfaces was made by means FDTD numerical simulations (see section 2.5).

3.2.1 Variation of the TMDC material

We work with the bulk and anisotropic materials of WS_2, MoS_2 and $MoSe_2$, since according to the simulations the anisotropic exhibited an interesting behavior, when compared to the hypothetical isotropic ones. To notice, although the refractive index might look the same for those materials, $MoSe_2$ refractive

index is quite different on the imaginary part since its absorption goes to 900nm wavelength, while MoS_2 and WS_2 is absorbing at 700nm [38], as it can be seen in the **Figure 1.8**.

Due to this factor, $MoSe_2$ will only start to show T after 900nm in the wavelength, which doesn't mean that the T values will be low, actually they were very interesting and compromising, so we studied what happens with more wavelength and more period, by running new simulations where we only changed the wavelength range and period. It is possible to see in the Table that the best material we found for these wavelengths was MoS_2, especially when adding SiO_2.

With the results from the simulations made by FDTD method and as expected of WS_2 and its properties, this was a pretty effective material to start with, mainly due to the high transmission values obtained.

3.2.2 Simulations of a grating of lines.

These simulations take long periods of time to run due to the mesh order, the steps (number of times a calculation is executed), of the filling factor and the processor power of the computer where the program is running.

The simulations of the first configurations were made for a period of 580nm, the TMDC thickness (t1) of 150nm and the metal thickness (tm) of 30nm. While for the second configuration we only added the SiO_2, with a thickness (t2) of 30nm, without changing the previous values of period, t1 and tm.

As we start to run those simulations, we notice a similar behavior on the results for the three materials when changing polarizations, there was high T for polarization 0 (horizontal direction or transverse-electric) and almost no T for polarization 90 (vertical direction or transverse-magnetic). This happened for both first and second configurations, on different wavelengths, different filling factors and for different thicknesses of TMDC. So, while making the plot line graphics of the simulations, we looked into the best conditions where we could possibly have a polarizer (to have high T for one polarization and almost no T for the other).

To summarize the large quantity results analyzed, based on the plot line graphics made by FDTD method, there are two tables that have this data, **Table 3.1** that corresponds to possible polarizers for the configuration with no silica, and **Table 3.2** that also has the same purpose but with SiO_2 on the structure. This tables have the corresponding wavelength (μm) to where it's possible to observe the best conditions for a polarizer (high T for the transverse-electric and low T for transverse-magnetic) in the infrared (IR). The sweep settings for the results obtained on the tables **3.1** and **3.2**, have 11 steps for

both ff and t1, with a mesh order of 3, except for MoS_2 of the second configuration where the sweeps ran with 21 steps for both ff and t1 and an increase of the mesh order value to 5.

This exception was due to the previous study, on which material had better T values for this wavelengths and conditions. Not only by implementing more steps on ff and t1, but also by increasing the mesh, we would obtain higher accuracy values of the simulation results, which would make the graphics look visually better and at the same time help us understand the best conditions to work with before proceeding to the next objectives.

3.3 Results

In the following **Figure 3.3**, it's possible to visualize the difference in how the T behaviors for WS_2, depending on the SiO_2 is deposited over the metal. If we look closely, there is a darker red on the graphic B for a 25 to 30nm t2 when compared to the graphic **A)**, that doesn't exhibit this dark red color for the same t2.

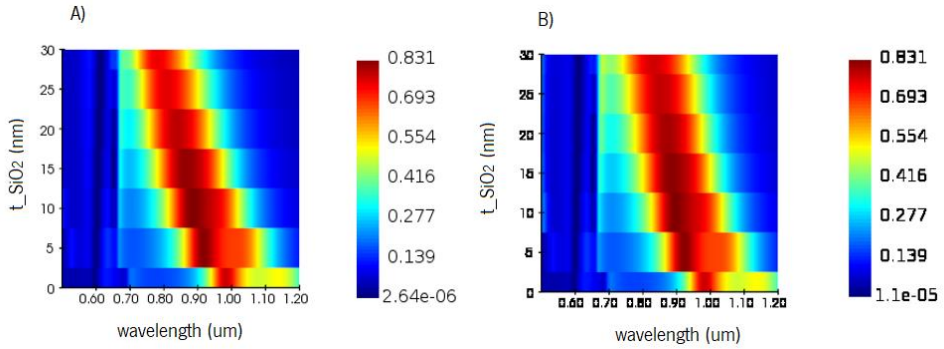


Figure 3.3: Comparison of two graphics with the same parameters for WS2, showing that it's better to work with SiO2 all over the metal. A) has less transmission in 25-30nm thickness; B) has more transmission for the same wavelength as graphic A) over the same parameters.

According to the obtained results, there was higher T for a thickness of 25 to 30nm when using SiO_2 all over the metal, which corresponds to the second configuration and graphic **B)** of the previous figure, compared to when the silica was deposited only below the TMDC. Also, for fabrication it was easier to work with the second configuration rather than making, at this small scales, such a precise deposition of silica that would need to be removed if in any case would happen to surpass the thickness of the TMDCs.

Table 3.1: Transmission and corresponding wavelength values obtained through FDTD numerical simulations for the materials WS2, MoS2 and MoSe2 for the first configuration (air, metal and TMDC).

	Filling factor	t1	Wavelength	Transmission
WS_2 (1 conf., pol: 0)	0.3	105nm	≈ 0.99 μm	≈ 0.78
WS_2 (1 conf., pol: 90)	0.3	105nm	≈ 0.99 μm	≈ 0.02
WS_2 (1 conf., pol: 0)	0.8	150nm	≈ 1.01 μm	≈ 0.74
WS_2 (1 conf., pol: 90)	0.8	150nm	≈ 1.01 μm	≈ 0.04
MoS_2 (1 conf., pol: 0)	0.3	150nm	≈ 1.08 μm	≈ 0.85
MoS_2 (1 conf., pol: 90)	0.3	150nm	≈ 1.08 μm	≈ 0.01
MoS_2 (1 conf., pol: 0)	0.8	135nm	≈ 1.06 μm	≈ 0.81
MoS_2 (1 conf., pol: 90)	0.8	135nm	≈ 1.06 μm	≈ 0.02
$MoSe_2$ (1 conf., pol: 0)	0.3	150nm	≈ 1.09 μm	≈ 0.76
$MoSe_2$ (1 conf., pol: 90)	0.3	150nm	≈ 1.09 μm	≈ 0.01
$MoSe_2$ (1 conf., pol: 0)	0.8	135nm	≈ 1.08 μm	≈ 0.68
$MoSe_2$ (1 conf., pol: 90)	0.8	135nm	≈ 1.08 μm	≈ 0.02

Table 3.2: Transmission and corresponding wavelength values obtained through FDTD numerical simulations for the materials WS2, MoS2 and MoSe2 for the second configuration (air, metal, SiO2 and TMDC).

	Filling factor	t1	t2	Wavelength	Transmission
WS_2 (2 conf., pol: 0)	0.3	105nm	25nm	≈ 0.86 μm	≈ 0.77
WS_2 (2 conf., pol: 90)	0.3	105nm	25nm	≈ 0.86 μm	≈ 0.008
WS_2 (2 conf., pol: 0)	0.8	60nm	25nm	≈ 0.92 μm	≈ 0.85
WS_2 (2 conf., pol: 90)	0.8	60nm	25nm	≈ 0.92 μm	≈ 0.007
MoS_2 (2 conf., pol: 0)	0.4	90nm	25nm	≈ 1.05 μm	≈ 0.63
MoS_2 (2 conf., pol: 90)	0.4	90nm	25nm	≈ 1.05 μm	≈ 0.005
MoS_2 (2 conf., pol: 0)	0.75	22.5nm	25nm	≈ 0.83 μm	≈ 0.84
MoS_2 (2 conf., pol: 90)	0.75	22.5nm	25nm	≈ 0.83 μm	≈ 0.39
$MoSe_2$ (2 conf., pol: 0)	0.3	75nm	25nm	≈ 0.88 μm	≈ 0.75
$MoSe_2$ (2 conf., pol: 90)	0.3	75nm	25nm	≈ 0.88 μm	≈ 0.006
$MoSe_2$ (2 conf., pol: 0)	0.8	90nm	25nm	≈ 0.91 μm	≈ 0.71
$MoSe_2$ (2 conf., pol: 90)	0.8	90nm	25nm	≈ 0.91 μm	≈ 0.007

3.3.1 Difference in the T after adding PDMS to the structure

The possibility to add a polymer to the system being the final made of: PDMS-TMDC-metal-TMDC-PDMS is useful for encapsulation. The PDMS envelopes the structure of the TMDC-metal-TMDC, so that it will be possible to manipulate and touch the structure without interfering directly (which would contaminate the samples) with the TMDC or the metal.

The results when adding PDMS to the structure, which has 150nm thickness of $MoSe_2$, 150nm of PDMS and 30nm of silver, shows no crucial difference over the T values, in the Figure the T for this configuration without PDMS (in purple color) reaches more or less 78% and with PDMS (in pink) achieves around 75% but in a different wavelength.

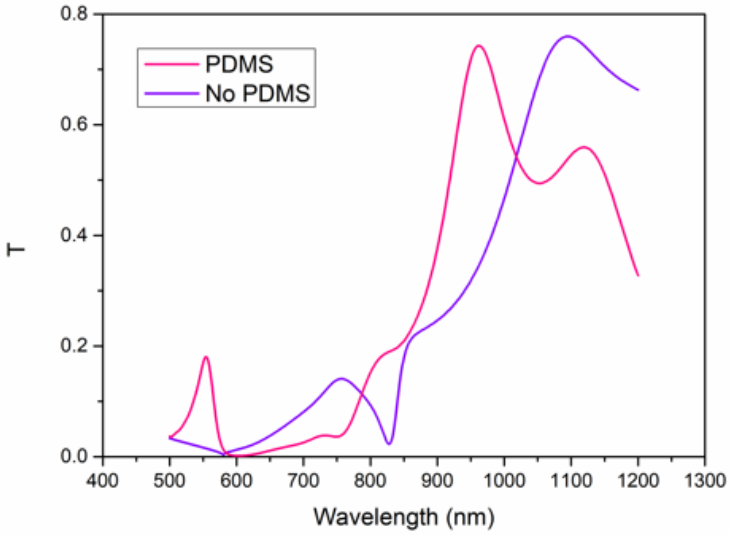


Figure 3.4: Transmission for first conf. of MoSe2 with and without PDMS on the structure.

In the fabrication process, it is expected that the PDMS in the structure won't interfere much with the transmission values.

3.3.2 Analysis of the angle variation

Having the previous simulations concluded, there was an urge to improve the characteristics of the results obtained in 2D and to have an almost flawless polarizer, so we checked the angles for all polarizations in order to have anti-reflection coating, which would make the transmission be more flat.

Applying the Snell's law in order to have refraction of the light, which is given by the equation: $n_{air}\sin(\theta_{air}) = n_{TMDC}\sin(\theta_{TMDC})$. The illustration in the **Figure 3.5**, is an example on how the light is refracted in the middles of our structure, where in red is the TMDC, light blue the air and in grey the metal.

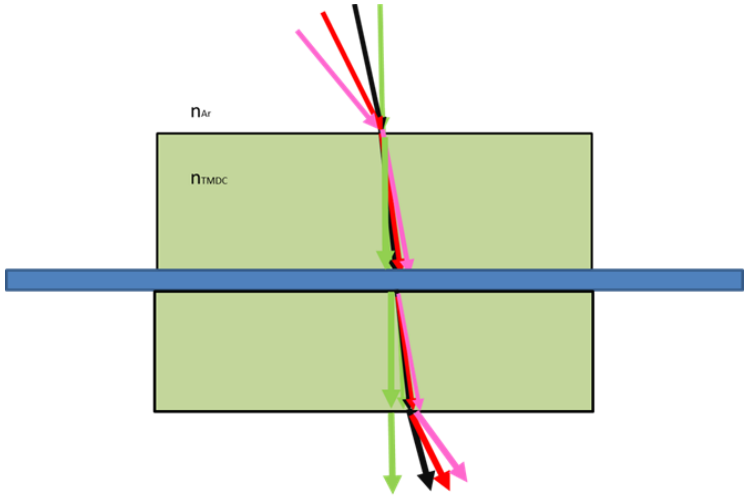


Figure 3.5: Illustration of the Snell's law in our structure for light refraction.

The graphics below, of the **Figures 3.6** and **3.7**, correspond to the T with the variation of the angle from 0 to 50 degrees (theta) in a wavelength range from 500 to 1200nm. In both of this figures, a) is when the polarization is 0 (transverse-electric) and b) the polarization is 90 (transverse-magnetic).

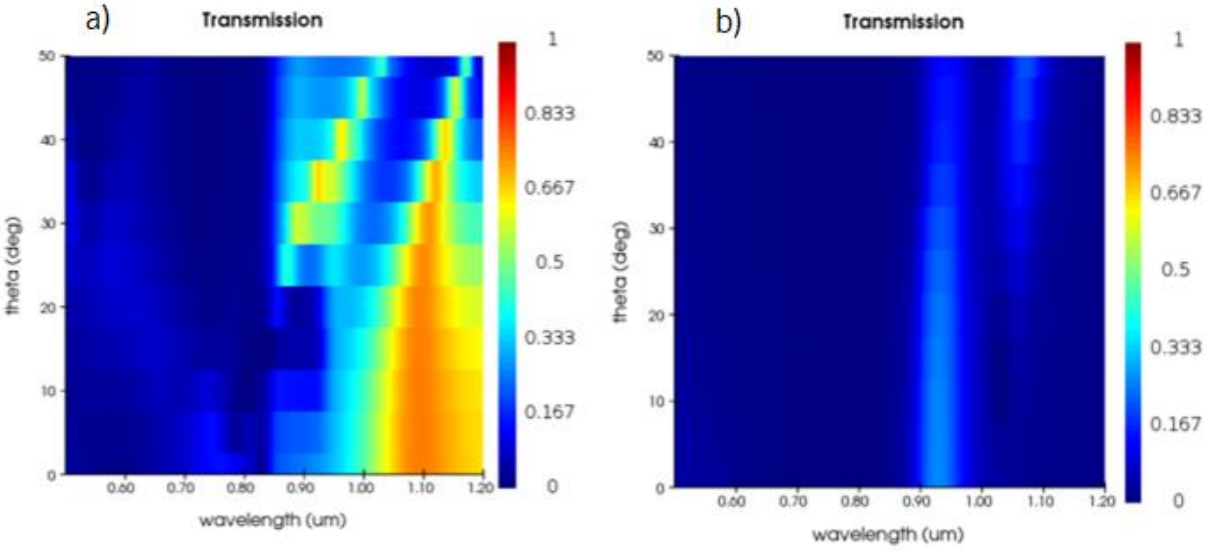


Figure 3.6: Transmission of MoSe2 (first conf.), where graphic a) is for polarization 0 and graphic b) for polarization 90.

In the **Figure 3.6**, which corresponds to the first configuration of $MoSe_2$, for a period of 500nm, a t_$MoSe_2$ of 150nm, the t_metal is 30nm and a filling factor of 0.3, proves to be a good polarizer in the IR range, due to in a) there is very narrow transmission from 0 to more or less 35 degrees and in b) no T for this angles.

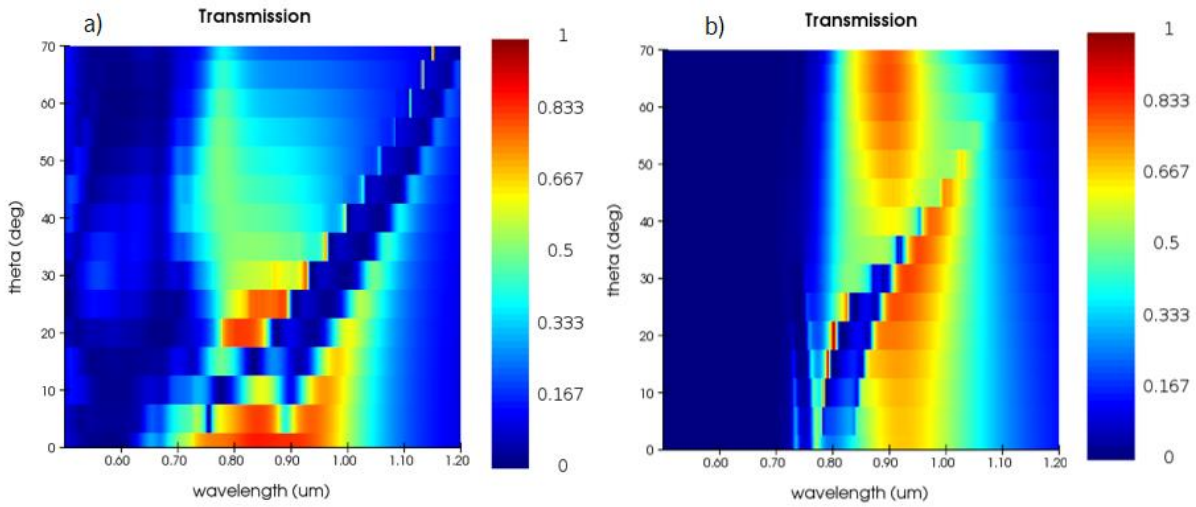


Figure 3.7: Transmission of MoS2 (second conf.) where graphic a) is for polarization 0 and graphic b) for polarization 90.

While in the **Figure 3.7**, corresponding to the second configuration of MoS_2, where the period is also 500nm, the t_MoS_2 is 22.5nm, t_metal of 30nm and a filling factor of 0.75, this material proves again to be the best for our system, since in both the polarizations there is T, but particularly in b) when the polarization is 90, there is T from 0 to 70 degrees,

3.3.3 Hexagonal or circular holes in the enhancement of the transmission by a square array

The results obtained for the T of the 3D plot graphics using the BFAST technique in Lumerical, with circular holes in the TMDC, for the second configuration of MoS_2 and the first configuration of $MoSe_2$, can be seen in the Figures below.

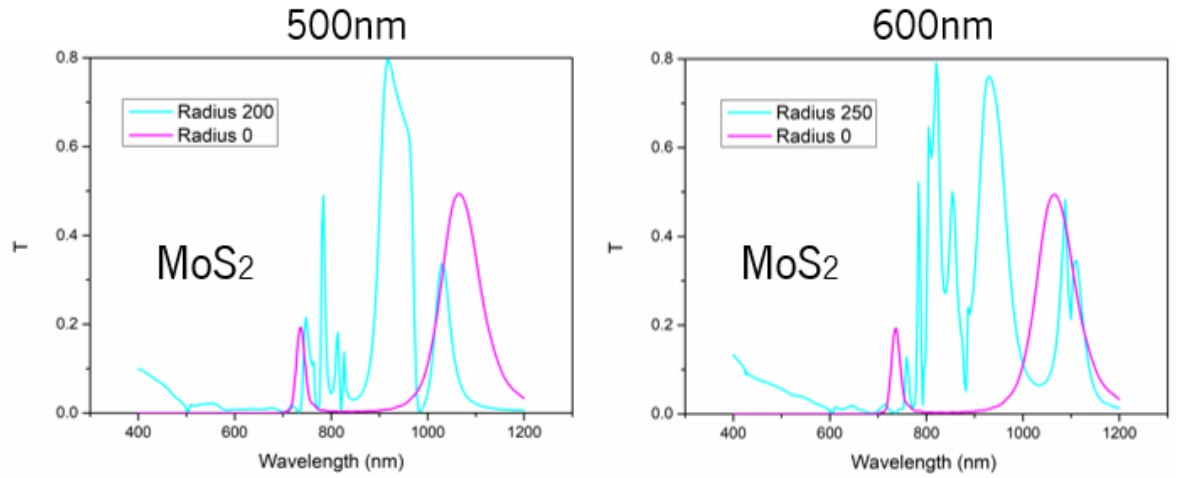


Figure 3.8: Transmission of the second configuration of MoS_2 for period 500 and 600nm, cyan color is when we have holes and pink for no holes.

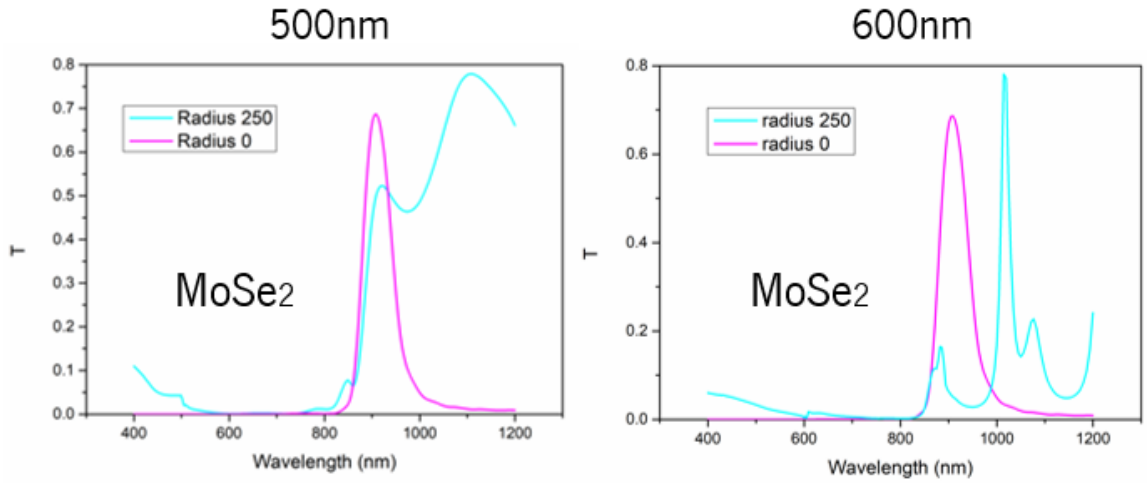


Figure 3.9: Transmission of the first configuration of $MoSe_2$ for period 500 and 600nm, cyan color is when we have holes and pink for no holes.

When the radius is 0 it means that the structure has no holes, which corresponds to the lines in pink, and the size of the holes (radius) can go from 0 to 500 or 600, corresponding to the periods of 500 or 600nm. The transmission is better for the first configuration of $MoSe_2$ with holes (which corresponds to the T in cyan color) and in the case of the second configuration of MoS_2 when we create holes to the structure the T increases from more or less 50% to 80%.

Finally, for the analysis of the holes in the TMDC, the results obtained of the two types of holes (circular or hexagonal) in the TMDC are very similar, with the exception of those two shown in the graphics below:

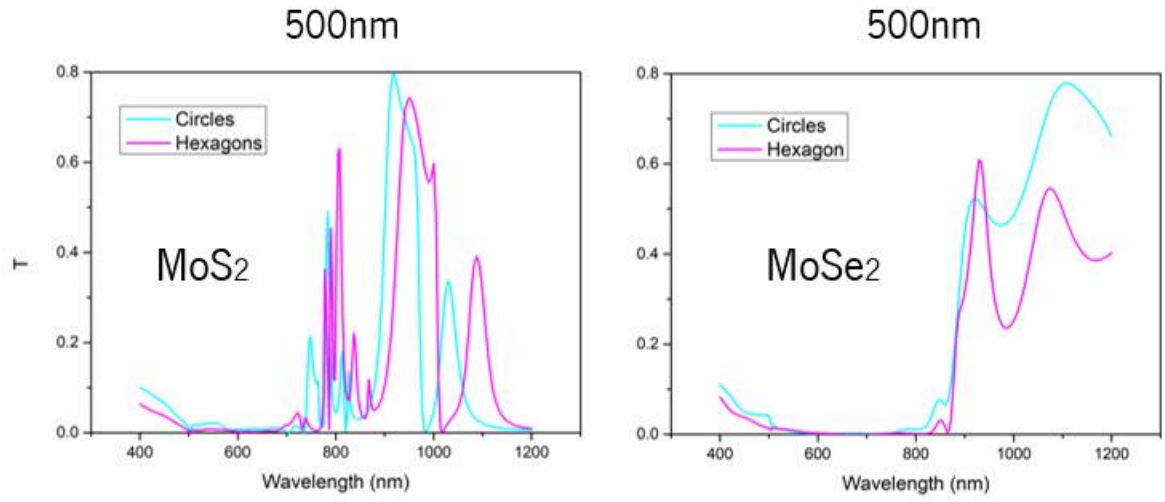


Figure 3.10: T obtained for second conf. of MoS_2 and first conf. of $MoSe_2$ for period 500nm, cyan color is for holes with circle structure and magenta for holes with hexagon structures.

As it can be seen in the **Figure 3.10**, it is better to make circular holes (cyan color) in the structure, instead of hexagons (pink color), for 500nm period of both the second configuration of MoS_2 and the first configuration of $MoSe_2$, but especially for the $MoSe_2$.

3.3.4 Transmission in the telecommunications range

The results for the telecommunications range are good, the T is around 40% to 50% for both configurations of MoS_2 and $MoSe_2$, but the T achieved in the graphic of the **Figure 3.11** stands out completely.

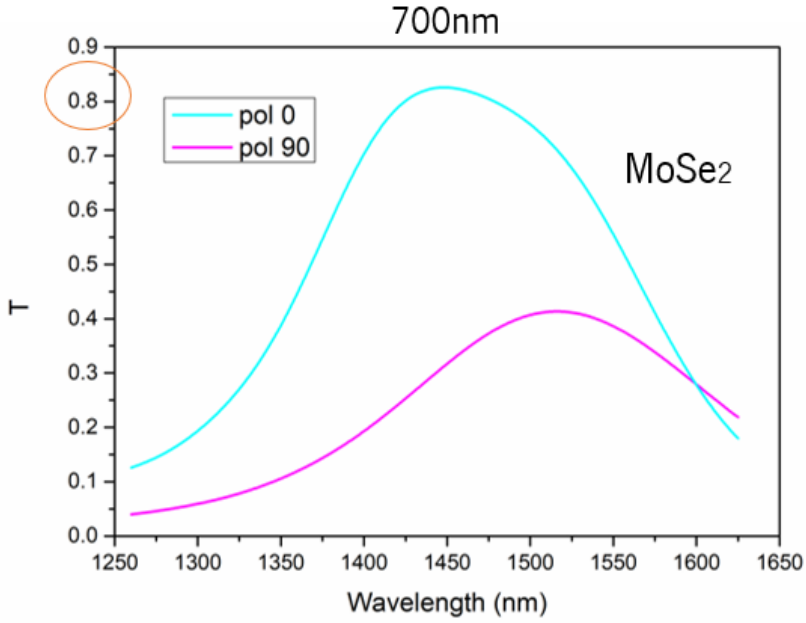


Figure 3.11: Transmission of the first configuration of $MoSe_2$ for period 700 t_$MoSe_2$ is 150nm, ff is 0.5, and t_metal is 30nm.

For a period of 700nm, and for both polarizations represented in cyan and pink colors, the structure achieves high and broadband transmission around 80% for one polarization and of 40% for the other. This makes our device suitable in the telecommunications range and the parameters to achieve this T are: t_$MoSe_2$ is 150nm, ff is 0.5, the t_metal is 30nm and the period 700nm.

3.3.5 Analysis of the mode resonance for high transmission

In the case of the first configuration of $MoSe_2$, as it can be seen in **Figure 3.12**, we have in **a)** anti-symmetric plasmons, corresponding to the highest T localized approximately 960nm in the wavelength and in **b)** symmetric plasmons, which penetrate less in the metal therefore they correspond to the lowest T in the 1100nm wavelength.

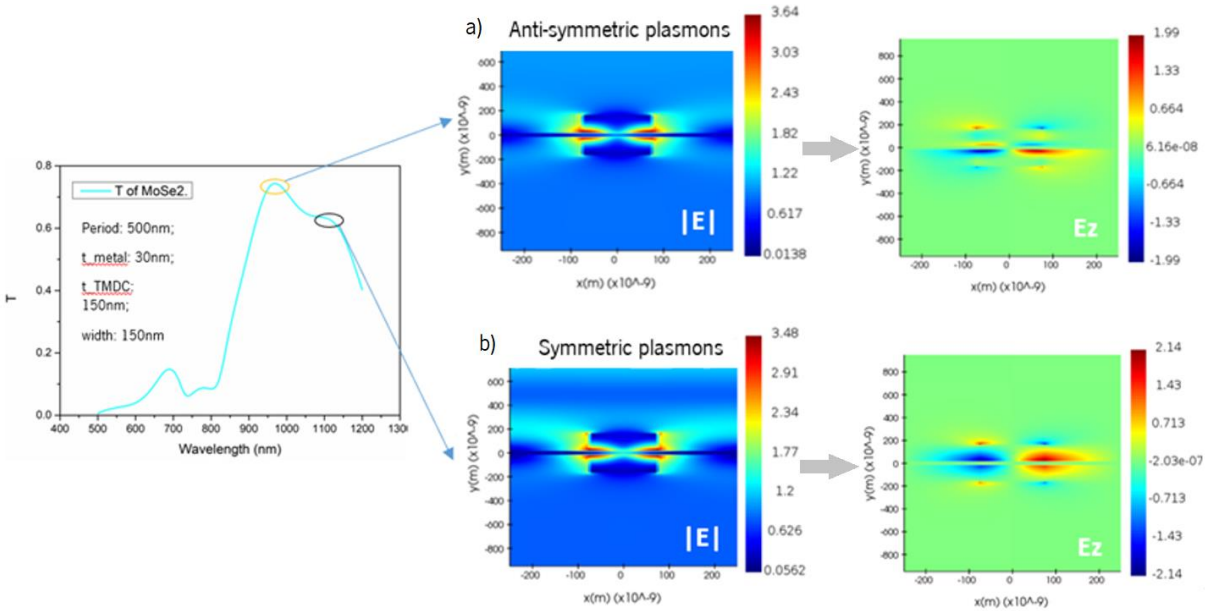


Figure 3.12: Analysis of the electric fields for the maxima of transmission of the configuration with the inset characteristics.

The high transmission comes from the gratings of the TMDC, and the electric fields are localized close to the metal (line in the middle in dark blue) because we want to create plasmons.

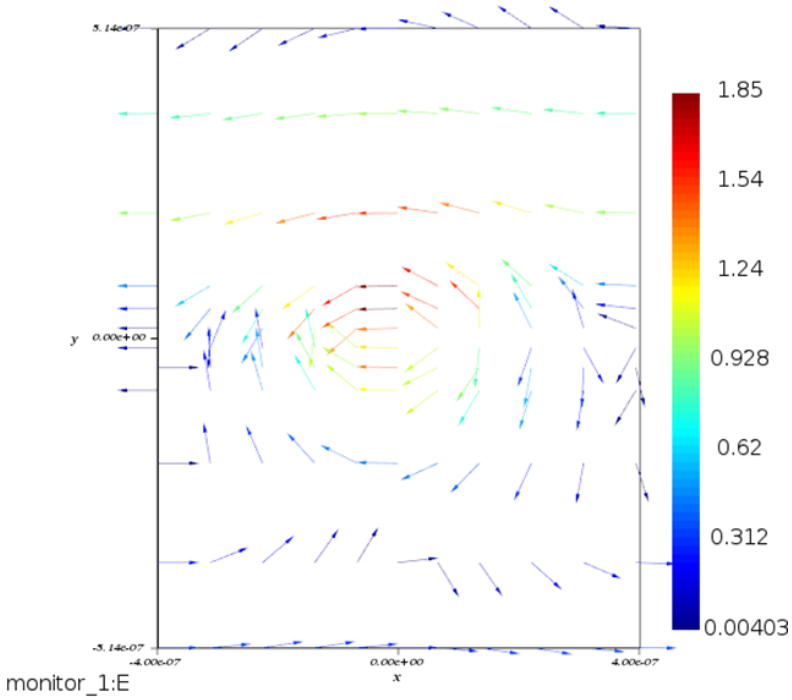


Figure 3.13: Movement of the electric field.

In the **Figure 3.13**, it is possible to visualize the movement of the electric field under the effect of the plasmonic resonance occuring in the thin metal layer, this phenomenon creates more T in the structure.

Chapter 4: FABRICATION OF A TRANSPARENT ELECTRODE

4.1 Introduction

As a proof of concept, we are going to use another configuration without nanopatterning and where we use gold instead of silver since it is more chemically stable, which makes the process of fabrication easier.

The structure of our device is TMDC-metal-TMDC and even though we made numerous simulations and studied the behavior of the structure with the angle variation, period optimizations, different thicknesses and widths for the metal and the TMDC, it was very difficult to control all the parameters in order to achieve the 90% T that we reached in the previous simulations.

Finally, in this chapter, we show the preliminary experimental studies of nanopatterning on flakes.

4.2 Materials and methods

So, we have in the **Figure 4.1** a scheme to help us visualize our structure which has air around the TMDC and the metal, although the air is not represented in the figure.

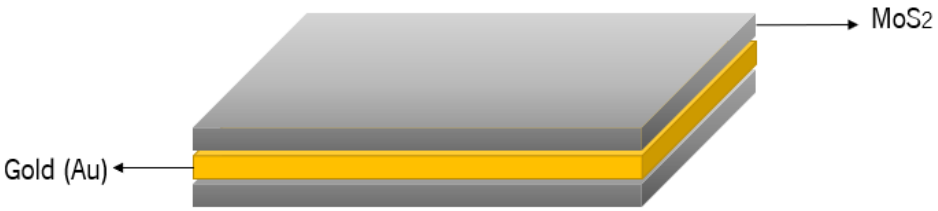


Figure 4.1: Illustration of the fabrication of the structure MoS2/Au/MoS2.

As a metal, the silver oxidizes a lot more in comparison to gold, for this we chose to fabricate with gold instead. So, we sputtered 30 nm of gold on a silicon wafer, and we measured its sheet resistance by means of the four probe point system (section 2.5) given a value R_s=5.2 Ω/sq.

It was not possible to have a precise control over the thickness of the exfoliated flakes from the sample of MoS_2 (from the **Figure 4.2**), only with Raman (position of the peaks and intensity) and the optical contrast we could study the different thicknesses of the flakes due to their different colors, as previously said in Chapter 2 in the optical characterizations.

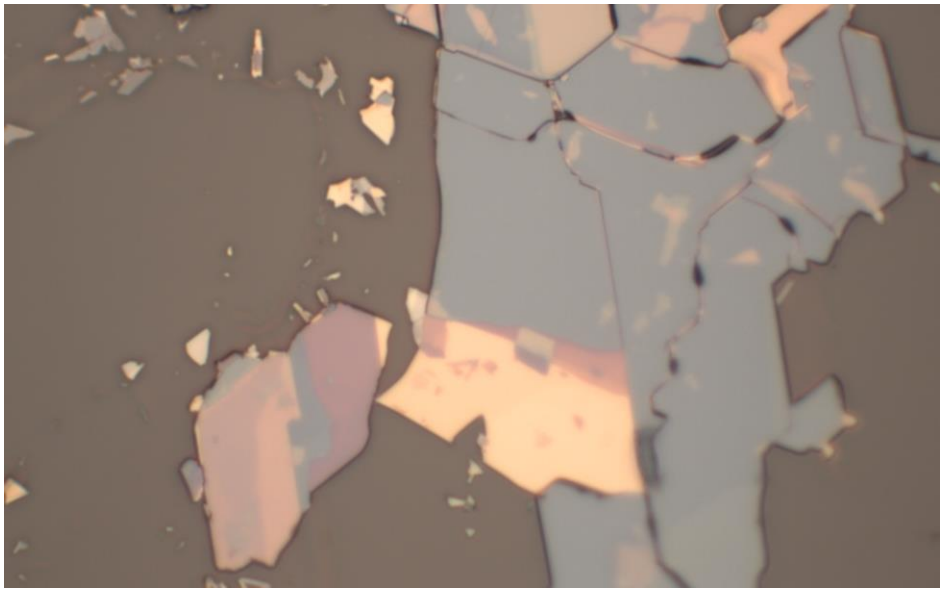


Figure 4.2: Different colors of the MoS2 flakes taken with the optical microscope.

In the **Figure 4.2** above, we have an example of the different colors of the MoS_2 flakes we observed in the optical microscope. It is possible to distinguish the blue flakes from the yellow and pink flakes quite easily.

The fabrication of our structure requires the following steps:

i. Exfoliation of a MoS_2 flake with the scotch tape technique, the exfoliated flakes can be seen in the **Figure 2.1**;

ii. Cut a small square of PDMS and press it on the thinnest exfoliated flakes that are in the tape.

iii. Gold was sputtered on silicon wafer without adhesion Cr layer, by Dr. Jerome Borme, can be seen in **Figure 4.3**. For easiness this wafer was divided in small sized 1cmx1cm square samples (which contain 30nm of gold) and finally, they were cleaned in an ultrasonic bath for 25 minutes at room temperature.



Figure 4.3: Tempered glass with 30nm of gold.

iv. Remove the square from the hot tube and dry them in compressed air;

v. Carefully press the thinnest flakes that we have from the second step on the gold thin film that is now cleaned and dried;

vi. The deposition of the flakes is made in a hot plate (the hot plate we use is in the microscope identified in the **Figure 4.4**) for one hour at 80° degrees;

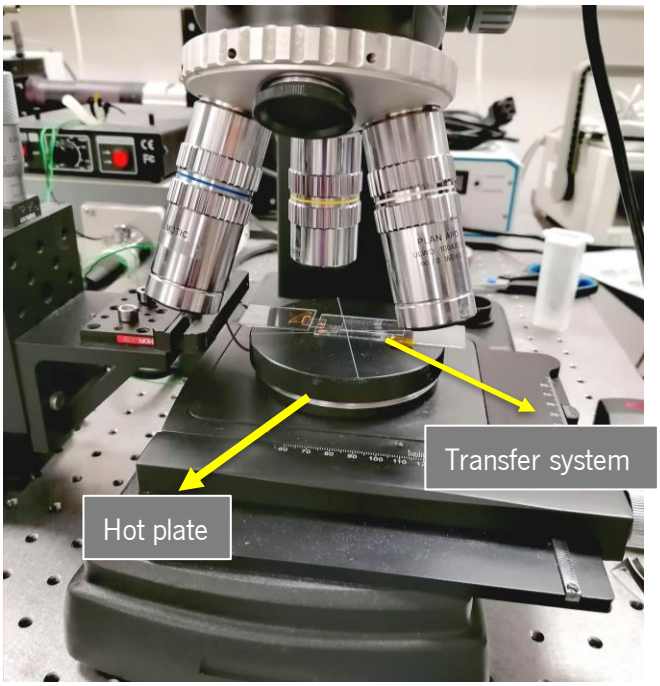


Figure 4.4: Microscope in which we make the system and also the deposition of the flakes in the hot plate.

vii. After this procedure, we peel of the gold together the PDMS and the flakes now visualize in the optical microscope and by Raman spectroscopy if our sample contains gold and MoS_2 flakes down the metallic film. (Raman spectrum shows less intensity peaks in the Au/MoS_2 than in the only MoS_2 flakes).

viii. Then, if the parameters at the step number seven are fulfilled, we make the analysis of the thicknesses of the MoS_2 in the AFM or profilometer;

ix. Repeat the same process from the first to the second step, but now the sample can only contain flakes of MoS_2;

x. Check in Raman and optical microscope if this sample contains small flakes of MoS_2;

xi. If the step number ten is fulfilled we place the coordinates in Raman, of where we have, which we consider to be the best, flakes of MoS_2 of the sample with gold and the sample without gold.

xii. The two samples are placed in different cover glasses and in order to make the transfer of the flake deposited in one sample to the other, we construct the system we have in the **Figure 4.5**, where **(A)** corresponds to the sample with PDMS, MoS_2 and gold, **(B)** the sample with MoS_2, **(C)** is also PDMS which helps in the transfer of the flakes from the bottom sample **(B)** to the sample

in the top **(A)** or the inverse, and finally **(D)** is a tape we use to stabilize the samples during the transfer.

By visualizing in the microscope, once we have the flakes precisely top and down we previously considered to be the best, we press very carefully on top of the sample **(A)** until we have the following order: PDMS-MoS_2-Au-MoS_2-PDMS-glass, in only one of the samples. The system can be manipulated in 3 coordinates x-y-z.

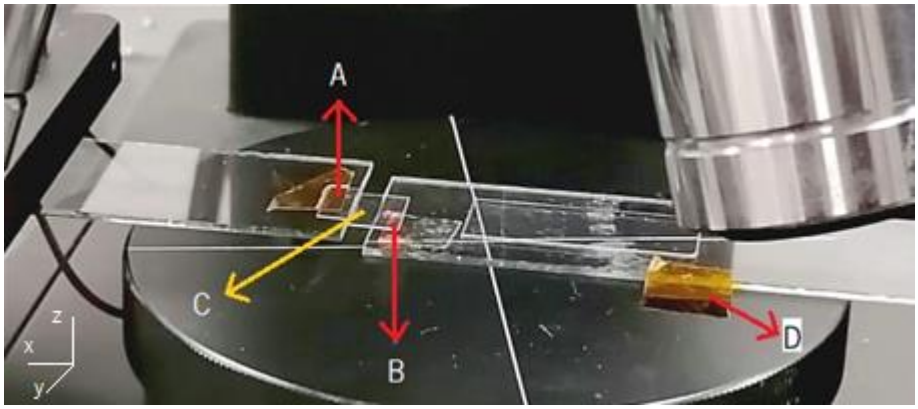


Figure 4.5: System used to make the transfer process of the flakes top-down, (A) sample with PDMS-MoS2-gold, (B) sample with MoS2, (C) is PDMS and (D) a tape that helps stabilizing the transfer system.

xiii. Check in Raman and the optical microscope if we have PDMS-MoS_2-gold-MoS_2-PDMS.

After we concluded this process several times, we had at our disposal ten samples, which we named of: sample 1, sample 2, etc.

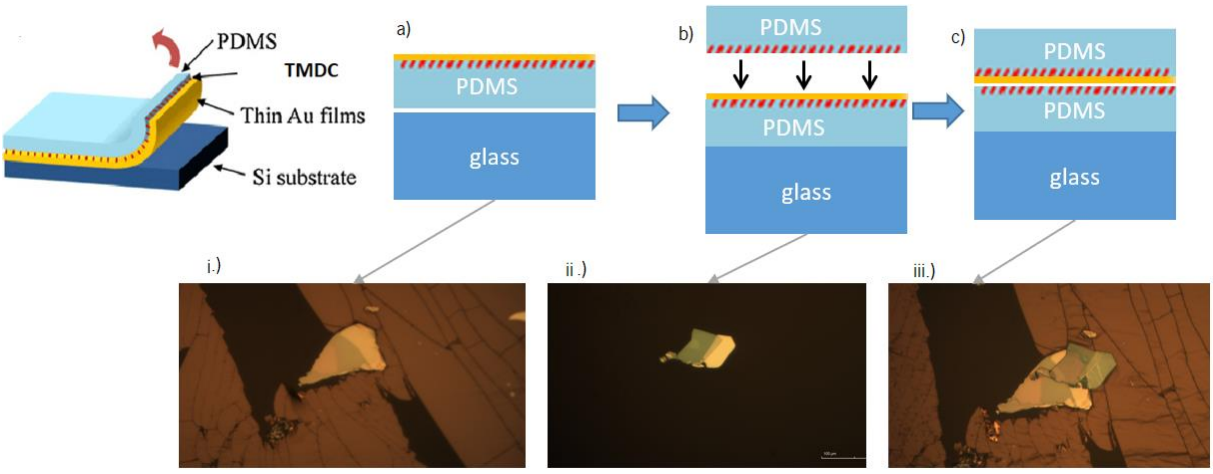


Figure 4.6: Illustration scheme of the fabrication process. i) bottom flake of MoS2 with Au on top in the upper right part. ii) top flake, iii) final structure PDMS-MoS2-gold-MoS2-PDMS.

To summarize, the scheme in the **Figure 4.6** explains, more simplified, the fabrication steps previously mentioned in how we managed to build this system. The peel-off of the PDMS with MoS_2 and

gold from the silicon substrate is placed in a) a glass slide with PDMS, TMDC and gold above it, that corresponds to **i)**.

Then we make the deposition of the other flake of TMDC and PDMS, which can be seen in **ii)**, where it's showing the second flake of MoS_2. After this deposition we obtain the final structure c), the PDMS-MoS_2-Au-MoS_2-PDMS, that we can observe in **iii)** the PDMS involving the two flakes of MoS_2 top-down the gold.

4.3 Results

The transmission of the samples was measured by FTIR (see section 2.3) in the range of 600nm to 2500nm for both the MoS_2-Au-MoS_2 structure and the Au. All the transmission measurements were normalized against 1cm thick cleaned glass and 2 layers of PDMS that were 250 μm thick, which we can consider transparent materials.

According to what was expected from the simulations and even though it was not possible to have a precise control over the thickness of the MoS_2 flakes, our structure enhances the T of the gold 6 times, as the T of the gold is less 10% in the wavelengths higher than 1400 nm and our device reaches 60% of transmission.

We show for space the best samples we obtain at our disposal are: sample 1, sample 3 and sample 6.

The flakes of the sample 1 (**Figure 4.7**) are a little thick but still, the structure achieves two high T peaks in 750 and 1070nm and two low T peaks in 860 and 1300nm. This sample is particularly good, especially if we require broadband T for a wavelength range from 1400 to 2400nm, in comparison with just having gold which exhibits practically no T for these wavelengths.

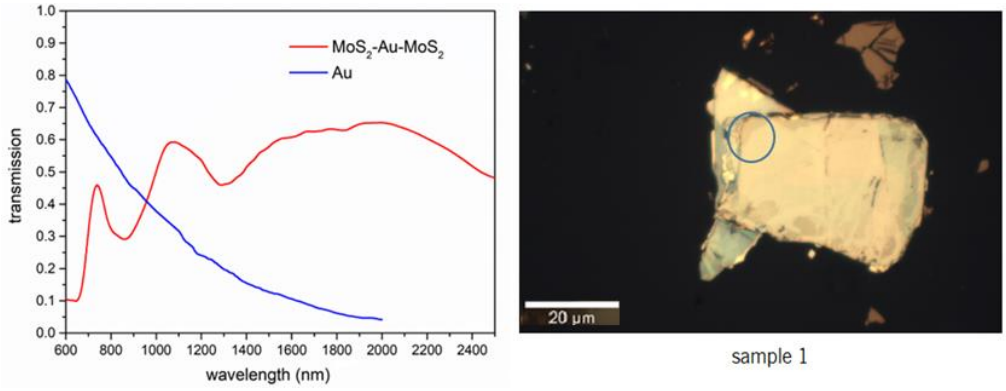


Figure 4.7: (left) Transmission spectra for MoS_2-Au-MoS_2 (red line) and Au (blue line). (right) Optical image of the measured structure of sample 1, the blue circle indicates the position of MoS_2-Au-MoS_2 layers.

Sample 3 reaches almost 70% of transmission even though we faced a challenge in making this samples, as previously said, it is very difficult to have a control over the thickness of the flakes and still this one reached a very high T in comparison with the gold, proving that if we improve and perfect our home-made system it can reach even higher transmission values.

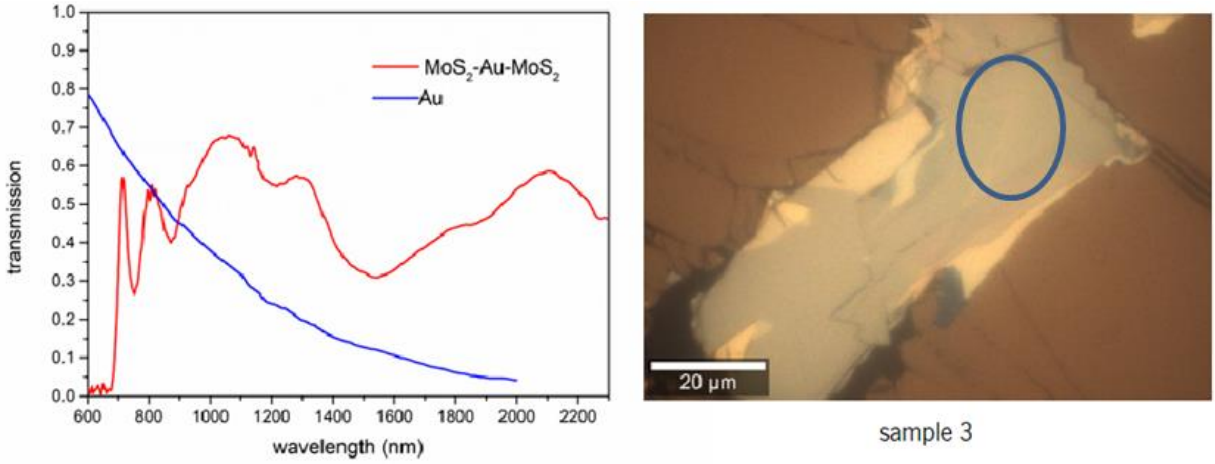


Figure 4.8: (left) Transmission spectra for MoS_2-Au-MoS_2 (red line) and Au (blue line). (right) Optical image of the measured structure of sample 3, the blue circle indicates the position of MoS_2-Au-MoS_2 layers.

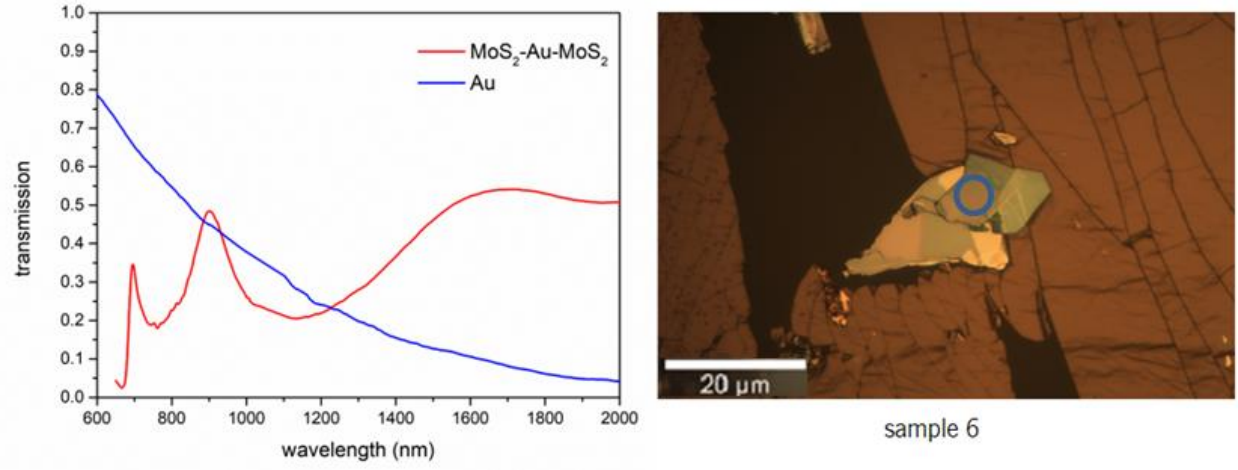


Figure 4.9:(left) Transmission spectra for MoS_2-Au-MoS_2 (red line) and Au (blue line). (right) Optical image of the measured structure of sample 6, the blue circle indicates the position of MoS_2-Au-MoS_2 layers.

Unfortunately, sample 6 does not have so much T like the other two samples, even show for higher wavelength show an enhancement of at least 5 times with respect the Au transmission, however due to the high thickness of the MoS_2 in this sample, it was possible to obtain this high quality pictures taken in the Ellipsometer as we see below.

Finally, we used the ellipsometer (see section 2.3) in order to verify the T values as a guarantee of the values given by FTIR in the working range of the ellipsometer from 700 to 1000 nm. Also, with this measurement we can obtain the transmission with direct sample visualization by contrast image.

In **Figures 4.10** and **4.11**, we show the enhancement of the transmission of MoS_2-Au-MoS_2 for sample 1 and 6, respectively, with respect the transmission of bare gold. Although the transmission of gold is lower in comparison with the FTIR measurements (this effect could be due to the local transmission it is easier not focus on the defects). We can observe that the minima and maxima of transmission for the metasurface fit very well for both measurements.

In the **Figures 4.10** and **4.11**, we also observe the direct visualization of transmission in sample 1 *and 6* where in II) the flakes are darker meaning that it has low T corresponding to the low peak in 850 nm (**Figure 4.10**) and *750nm* (**Figure 4.10**) wavelength and I) the flakes are brighter which corresponds to the 750 nm (**Figure 4.11**) and *900nm* (**Figure 4.11**) in the wavelength. In this figures we can identify the air around the structure, due to its contrast difference because and also observe the strip of peel off gold between the flakes of MoS_2.

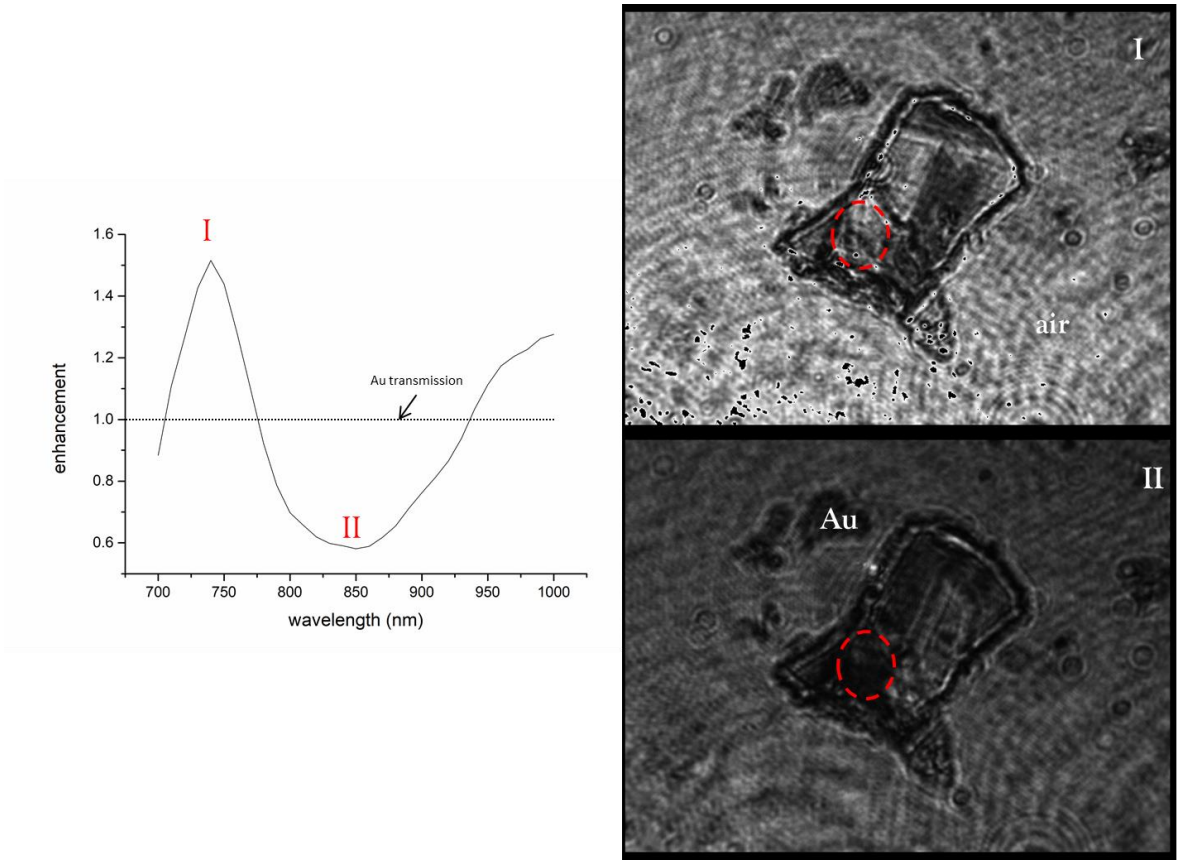


Figure 4.10: (left) Transmission spectra for MoS2-Au-MoS2 (red line) and Au (blue line). (right) Optical image of the measured structure the red circle indicates the position of MoS2-Au-MoS2 layers. Sample 1.

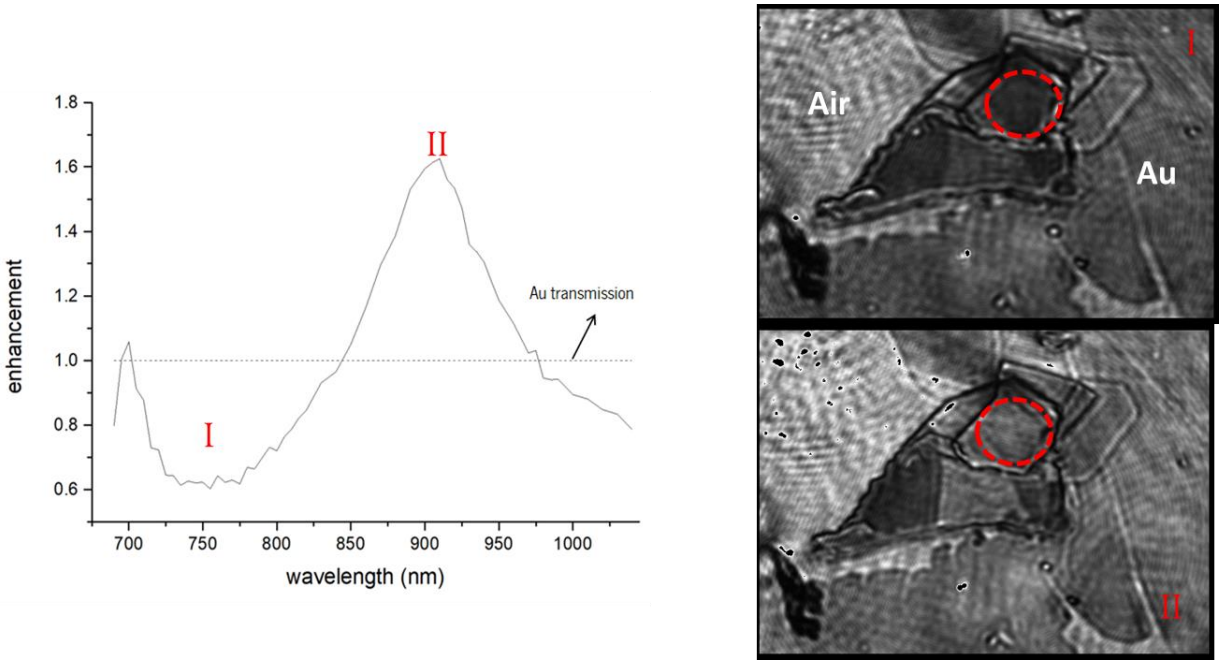


Figure 4.11: 4.11(left) Transmission spectra for MoS2-Au-MoS2 (red line) and Au (blue line). (right) Optical image of the measured structure the red circle indicates the position of MoS2-Au-MoS2 layers. Sample 6.

4.4 Preliminary results in nanopatterning of MoS_2

It was possible to reach out to a conclusion about the thickness of the flakes with the help of the lines made in Raman laser, which can be seen in the **Figure 4.12** The power of the laser could go from 0 to 30mW.

The lines, which look smooth and flawless in the pink flake (**Figure 4.12 a)**), were made with a laser power of 17.2mW and in **b)** the measurement of their length (more or less 9660nm, width around 403nm and the distance 2160nm between them.

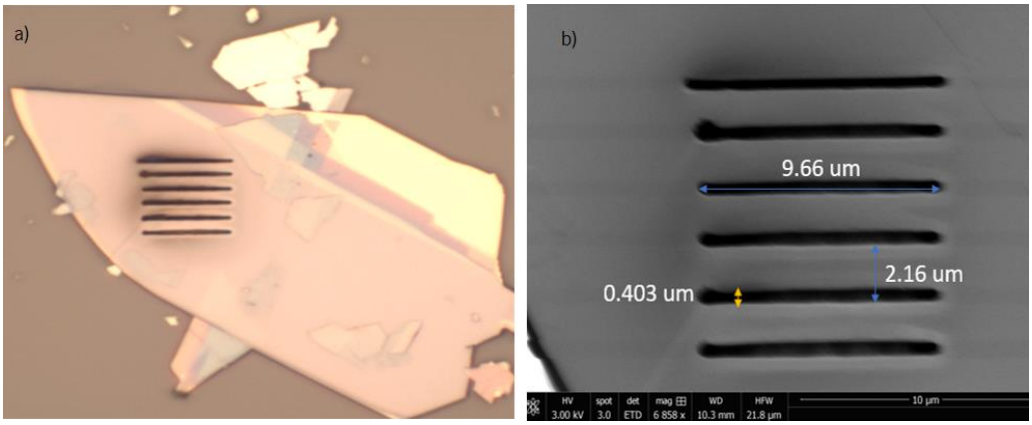


Figure 4.12: (a) optical microscope of a grating made by laser ablation on MoS2. (b) SEM image.

With 17mW laser power was not possible to make lines in the yellow and blue flakes.

The lines that can be seen in **Figure 4.13** made in the yellow flake for a laser power of 25mW in **a)** look very irregular and distorted and it was not possible to perfect them, by decreasing or increasing the power and even after reprograming the codes that make this lines it was still not possible to get a better result.

The blue flake in **b)** in comparison to the pink and yellow flakes is the most thick, not even with 30mW power was possible to make lines in the flakes. We tried with other blue flakes, but all exhibited the same behavior, the system requires more laser power to make lines in such thickness.

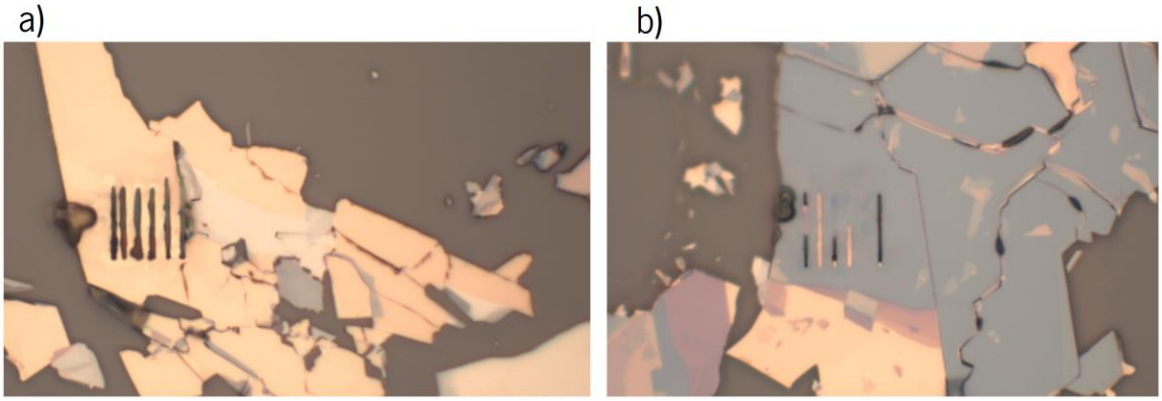


Figure 4.13: Attempts of laser ablation with thicker flakes.

With the lines, we conclude that the flakes with less thickness are pink, the ones that follow are yellow and finally the flakes with more thickness are the blues.

Finally, we also try to make holes In order to see the main geometry that we can obtain, in this way we obtain circular holes as it is shown in **Figure 4.14 (a)** and hexagonal holes (**Figure 4.14 (b)**). These results are related with the designs of a square array of holes in section 3.3.5.

Future work is needed to make the final nanostructure, maybe optimizing the laser ablation or choosing other etching process.

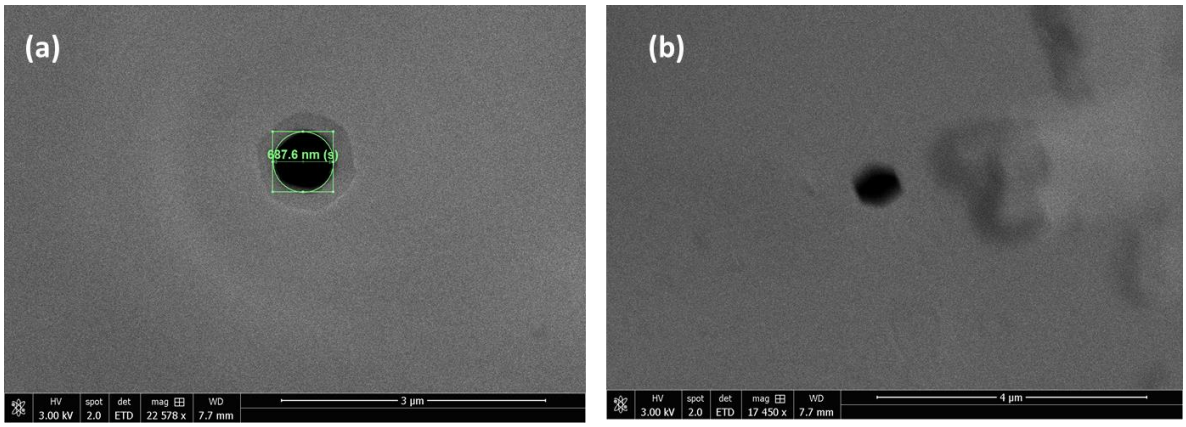


Figure 4.14: (a) circular and (b) hexagonal hole made by laser ablation on MoS2.

Chapter 5: DISCUSSION AND RESULTS

In this chapter we try to summarize the conclusions of the work.

- We achieved maximum values of transmission of 90%.for the second configuration (**Figure 5.1**) of Mos_2.

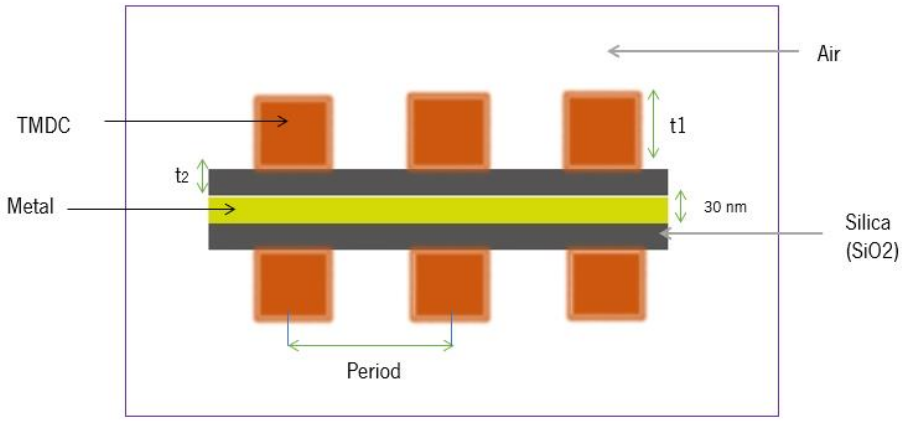


Figure 5.1: Illustration of the layout structure for the 2nd configuration, where t1 is the thickness of the TMDC and t2 the thickness of the SiO2.

- Better T values for lower thicknesses of the TMDCs.
- In the 1st configuration, changing the material (anisotropic to isotropic) had no effect over the T and both polarizations.
- Possible to visualize the impact of the plasmonic effect over the T.
- Adding PDMS to the structure doesn't critically affect the transmission values.
- The first configuration of $MoSe_2$ and for a period of 700nm, our structure achieves high and broadband T around 80% for one polarization and of 40% for the other, which can be used in the telecommunications range.
- Good transmission values until 70 degrees, for the second configuration of MoS_2.
- Since $MoSe_2$ has such a high absorption, it only shows T for wavelengths higher than 900nm.

- In the **Figure 5.2**, we have in **A)** the enhancement of adding the TMDC above and below the thin metal layer, in **B)** the very high and broadband T of MoS_2, which we conclude to be the best material to use in our structure and finally, since MoS_2 proves to be the most suitable material to use then the graphic **C)** corresponds to the second configuration of MoS_2 with T for one polarization (transverse-electric in blue color) and very low T for the other (transverse-magnetic the black line) which can be suitable as a polarizer in the infrared range.

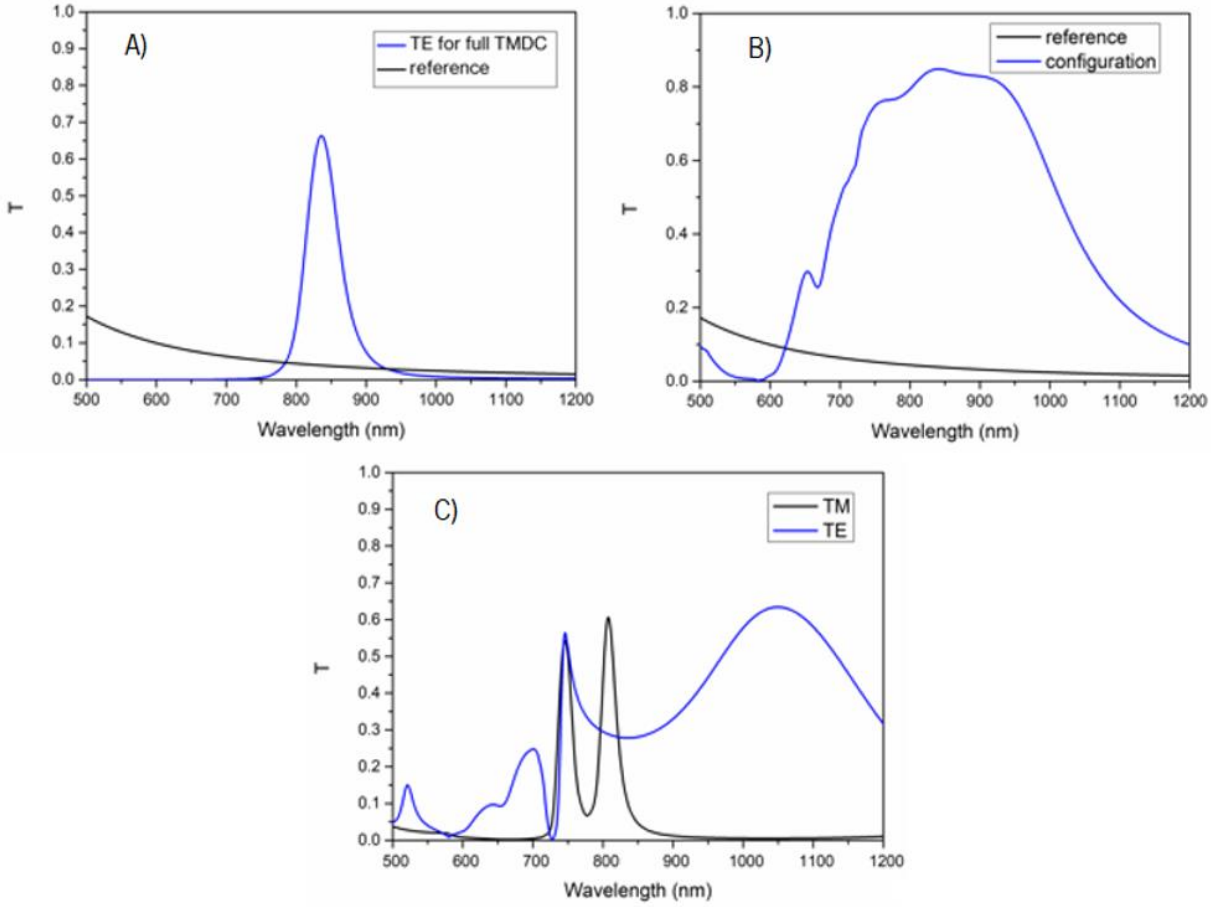


Figure 5.2: (A) In black is the transmission of when we just have metal and in blue is after we add a layer of TMDC top-down the metal, (B) High and broadband transmission around 90% and (C) Suitable polarizer in the IR.

- As a proof of concept, we fabricated and measured a system composed of PDMS-MoS_2-Au-MoS_2-PDMS obtaining values of transmission of 60% in a broadband range of wavelengths in the near infrared and peaks of 70% of transmission for a 30 nm thick gold layer and a sheet resistance of 5 Ω/sq.

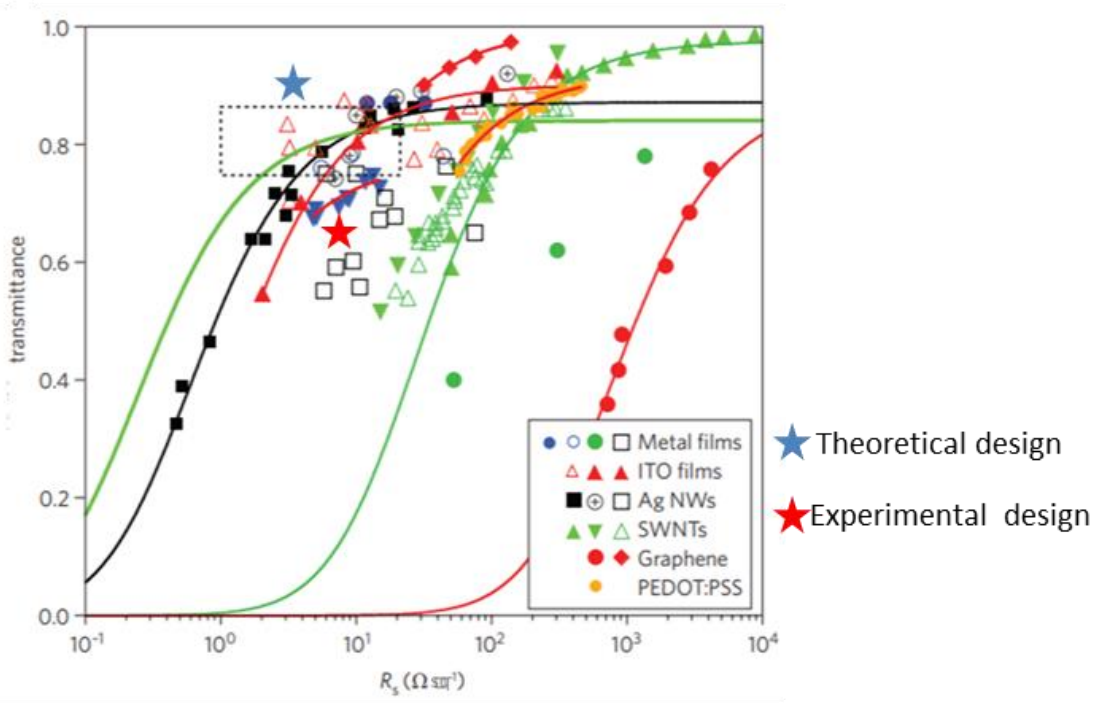


Figure 5.3: Our work with respect other electrodes (Modified from [38]).

Chapter 6: CONCLUSIONS AND FUTURE WORK

To summarize in **Figure 5.3**, we add to the values of different transparent electrodes our theoretical and experimental system, showing that ours have a very good performance with respect to other electrodes, even considering that the experimental design is only a proof of concept.

We thought of the possibility to measure, in the future, the transmission of the samples with angle variation in the Ellipsometer, because by numerical simulations we think that the sample is robust with angle as we see in section 3.3.2.

Also, the optimization of the laser ablation and maybe using other etching process, to make the final nanostructure and finally, measurements of the thickness on different parts of the samples when the AFM is available, due to the fact that it was not operative during the time of this thesis.

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